Living in the IT Era

Mayra Christina M. Ambrocio, DEM
Roel L. Fucio
Milben A. Bragais
Ozzy Boy Nicopior
Michael T. Ardizone

College of Arts and Sciences

Table of Contents

Module 5. Networks And Types
Introduction 81
Learning Objectives 81
Lesson 1. Definition of Terms 82
Lesson 2. Types: LAN, MAN, WAN, VPN 84
Lesson 3. Components of a Network 85
Lesson 4. E-Commerce 96
Lesson 5. Internet or the Internetwork 111
Lesson 6. Ips, Protocols, Browsers, URL 117
Lesson 7. ISPs 126
Lesson 8. Ethics or Netiquette 128
Lesson 9. Issues and Problems 130

Module 6. Data Seurity, Data Privacy And Protection
Introduction 140
Learning Objectives 140
Lesson 1. Computer and Data Security 141
Lesson 2. Definitions and Views on Data 145
Lesson 3. Data Privacy Act 146
Lesson 4. Cases Examples, Issues and Concerns 150

Module 7. Political, Economic and Societal Impacts and Change
Introduction 157
Learning Objectives 157
Lesson 1. Economic Impacts of ICT 158
Lesson 2. Political Impacts of ICT 160
Lesson 3. Social Impacts of ICT 162

List of Figures

Figure	Description	Page
5.1	Local Area Network (LAN) Setup	85
5.2	Metropolitan Area Network (MAN) Setup	87
5.3	Wide Area Network (WAN) Setup	88
5.4	Virtual Private Network (VPN) Setup	89
5.5	A stand-alone NIC used in a Desktop Computer	96
5.6	Hub Setup	97
5.7	Switch Setup	98
5.8	Router Setup	99
5.9	Modem Setup	100
5.10	Network Cables and Connectors	102
5.11	Google Interface using Internet Explorer	114

List of Tables

Table	Description	Page
5.1	Definition of Network	82
5.2	Advantages of E-Commerce	109
5.3	Disadvantages of E-Commerce	109
5.4	Parts of Internet Explorer	114

MODULE 5
NETWORKS AND TYPES

Introduction

Emerging technology used for everything from accessing the internet or printing a document to downloading an attachment from an email, networks are the backbone of business today. They can refer to a small handful of devices within a single room to millions of devices spread across the entire globe and can be defined based on purpose and/or size (Cloud, 2015).

There are various types of computer networks available. We can categorize them according to their size as well as their purpose. The size of a network should be expressed by the geographic area and number of computers, which are a part of their networks. It includes devices housed in a single room to millions of devices spread across the world (Bourgeois, 2016).

One way to categorize the different types of computer network designs is by the scope or scale of the network. For historical reasons, the networking industry refers to nearly every type of design as some type of area network.

Learning Outcomes

At the end of this module, students should be able to:

1. Define the meaning and nature of Networks and its types.
2. Identify the importance of Networks.
3. Enumerate the advantages and disadvantages of Networks.
4. Differentiate each Networks' Types.
5. Demonstrates understanding of Networks and its' Types.

Lesson 1. Definition of Terms

A Brief History

The first computer networks were links between physical workstations, personal and desktop computers either linked by Ethernet cabling, or later, wirelessly connected. Each computer had its own physical hard drive, and the hard drives of the computers were often represented by different drive letters on the workstation's operating system interface and different kinds of software applications. A user could go onto a physical computer with its own drive, pull from the computer's drive, or log onto network software to access the drive of a different computer in a different part of a room or building (Cloud, 2015).

In these early setups where end-users often selected access to a networked drive from a drop-down list, engineers used specific network topologies to link these workstation components together.

Hub, ring, bus, star and tree topologies were some of the common networking structures used to allow the individual computers to "talk" to one another and share information. The computers were usually co-located in a particular building but perhaps networked throughout various rooms or offices.

Later, the virtual network emerged along with other virtualized computing processes. This happened in conjunction with the practice of remote networking, as the cloud and other innovations promoted wide-area network designs (Cloud, 2015).

Table 5.1 Definition of Network

TERMINOLOGY	DEFINITION
Network	- Is a collection of computers, servers, mainframes, network devices, peripherals, or other devices connected to one another to allow the sharing of data? An example of a network is the Internet, which connects millions of people all over the world. To the right is an example image of a home network with multiple computers and other

	network devices all connected (ComputerHope, 2020). - A fabric or structure of cords or wires that cross at regular intervals and are knotted or secured at the crossings (Addison, 1990). - A network consists of two or more computers that are linked in order to share resources (such as printers and CDs), exchange files, or allow electronic communications. The computers on a network may be linked through cables, telephone lines, radio waves, satellites, or infrared light beams (Winkelman, 2013). - A computer network, also referred to as a data network, is a series of interconnected nodes that can transmit, receive and exchange data, voice and video traffic (Gillis, 2019). - A network is defined as a group of two or more computer systems linked together (Beal, 2020)

The network connects computers for the purpose of a. Sharing information, devices like printers/scanners and applications b. User Collaboration through Email, Chats and Video conferencing c. playing games d. Voice calls. Computers are connected through network devices like Hub, Switch, Routers, Network Interface Card (NIC) and transmission media like wired connections (coaxial cables, optical fiber) and wireless (Antenna). The network is the backbone of the IT evolution and in the digital world, it plays an important role in connecting everything on the earth like equipment, machines, appliances, gadgets and moving objects. Networks can be classified based on its size and purpose it serves. The size of the network covers the geographical spread and the volume of the computers connected. In this article let us analyze several types of networks (Cloud, 2015).

A network, in computing, is a group of two or more devices or nodes that can communicate. The devices or nodes in question can be connected by physical or wireless connections. The key is that there are at least two separate components, and they are connected (Cloud, 2015).

The scale of a network can range from a single pair of devices or nodes sending data back and forth, to massive data canters and even the global Internet, the largest network in existence. What all of these networks have in common, from the smallest ones to the largest, is that they allow computers and/or users to share information and resources. Networks may be used for:

- Communications such as email, instant messaging, chat rooms, etc.
- Shared hardware such as printers and input devices.
- Shared data and information through the use of shared storage devices.
- Shared software, which is achieved by running applications on remote computers.

Lesson 2. Types: LAN, MAN, WAN, VPN

The size of the Network can vary from connecting two computers inside a small room to lakhs of computers across the world. Different types of networks are:

What is Local Area Network (LAN)?

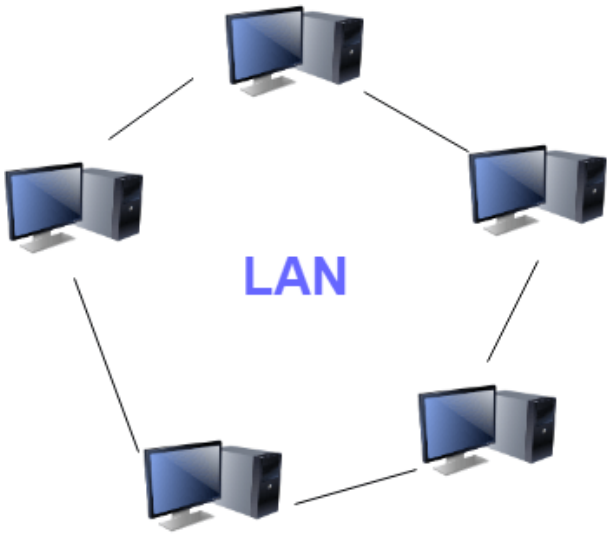

Figure 5.1 Local Area Network (LAN) Setup

Source: https://www.guru99.com/types-of-computer-network.html

A *Local Area Network (LAN)* is a group of computer and peripheral devices which are connected in a limited area such as school, laboratory, home, and office building. It is a widely useful network for sharing resources like files, printers, games, and other application. The simplest type of LAN network is to connect computers and a printer in someone's home or office. In general, LAN will be used as one type of transmission medium. It is a network which consists of less than 5000 interconnected devices across several buildings (Deeksha, 2020).

Characteristics of Local Area Network (LAN) (Deeksha, 2020)

Here are important characteristics of a LAN network:

- ✓ It is a private network, so an outside regulatory body never controls it.
- ✓ LAN operates at a relatively higher speed compared to other WAN systems.
- ✓ There are various kinds of media access control methods like token ring and ethernet.

Advantages of LAN (Deeksha, 2020)

Here are pros/benefits of using LAN:

- ✓ Computer resources like hard-disks, DVD-ROM, and printers can share local area networks. This significantly reduces the cost of hardware purchases.
- ✓ You can use the same software over the network instead of purchasing the licensed software for each client in the network.
- ✓ Data of all network users can be stored on a single hard disk of the server computer.
- ✓ You can easily transfer data and messages over networked computers.
- ✓ It will be easy to manage data at only one place, which makes data more secure.
- ✓ Local Area Network offers the facility to share a single internet connection among all the LAN users.

Disadvantages of LAN (Deeksha, 2020)

Here are the important cons/ drawbacks of LAN:

- ✓ LAN will indeed save cost because of shared computer resources, but the initial cost of installing Local Area Networks is quite high.
- ✓ The LAN admin can check personal data files of every LAN user, so it does not offer good privacy.
- ✓ Unauthorized users can access critical data of an organization in case LAN admin is not able to secure centralized data repository.
- ✓ Local Area Network requires a constant LAN administration as there are issues related to software setup and hardware failures.

What is Metropolitan Area Network (MAN)?

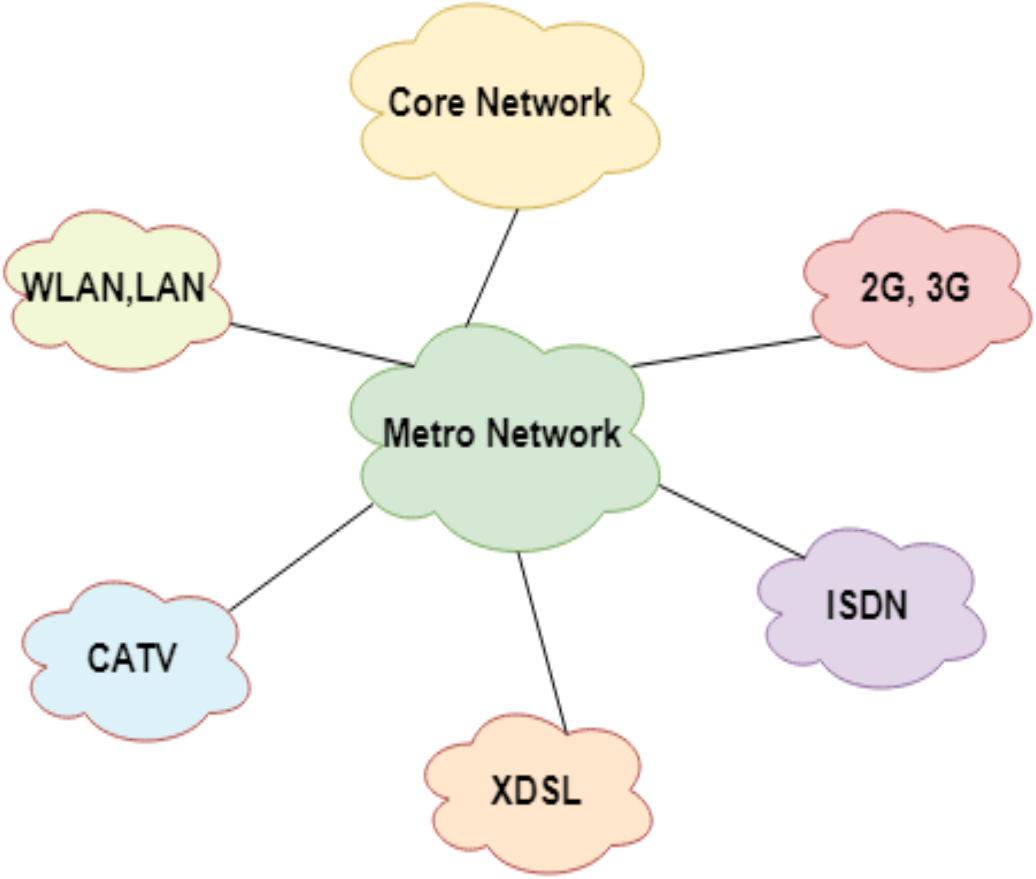

Figure 5.2 Metropolitan Area Network (MAN) Setup

https://www.guru99.com/types-of-computer-network.html

A *Metropolitan Area Network (MAN)* is consisting of a computer network across an entire city, college campus, or a small region. This type of network is large than a LAN, which is mostly limited to a single building or site. Depending upon the type of configuration, this type of network allows you to cover an area from several miles to tens of miles (Deeksha, 2020).

Characteristics of Metropolitan Area Network (MAN) (Deeksha, 2020)

Here are important characteristics of the MAN network:

- ✓ It mostly covers towns and cities in a maximum 50 km range
- ✓ Mostly used medium is optical fibers, cables
- ✓ Data rates adequate for distributed computing applications.

Advantages of MAN (Deeksha, 2020)

Here are pros/benefits of using MAN system:

- ✓ It offers fast communication using high-speed carriers, like fiber optic cables.
- ✓ It provides excellent support for an extensive size network and greater access to WANs.
- ✓ The dual bus in MAN network provides support to transmit data in both directions concurrently.
- ✓ A MAN network mostly includes some areas of a city or an entire city.

Disadvantages of MAN (Deeksha, 2020)

Here are drawbacks/cons of using the MAN network:

- ✓ You need more cable to establish MAN connection from one place to another.
- ✓ In MAN network it is tough to make the system secure from hackers

What is Wide Area Network?

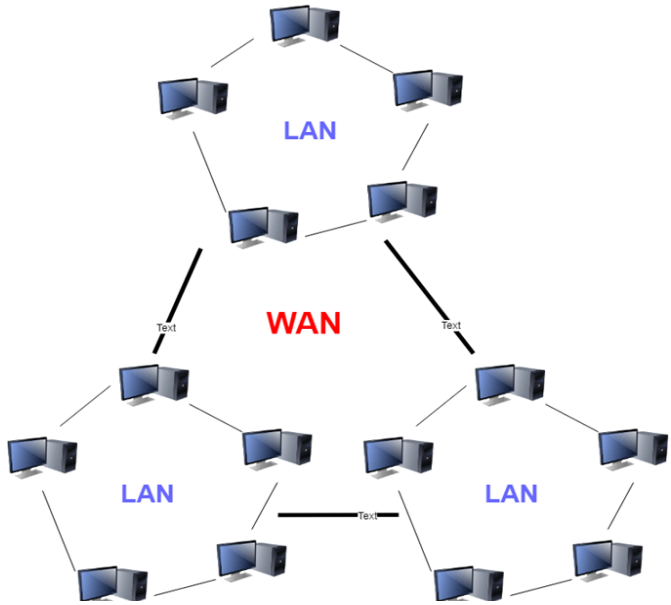

Figure 5.3 Wide Area Network (WAN) Setup

https://www.guru99.com/types-of-computer-network.html

WAN (Wide Area Network) is another important computer network that which is spread across a large geographical area. WAN network system could be a connection of a LAN which connects with other LAN's using telephone lines and radio waves. It is mostly limited to an enterprise or an organization (Deeksha, 2020).

Characteristics of LAN: (Deeksha, 2020)

- ✓ The software files will be shared among all the users; therefore, all can access to the latest files.
- ✓ Any organization can form its global integrated network using WAN.

Advantages of WAN (Deeksha, 2020)

Here are the benefits/ pros of using WAN:

- ✓ WAN helps you to cover a larger geographical area. Therefore, business offices situated at longer distances can easily communicate.
- ✓ Contains devices like mobile phones, laptop, tablet, computers, gaming consoles, etc.
- ✓ WLAN connections work using radio transmitters and receivers built into client devices.

Disadvantage of WAN (Deeksha, 2020)

Here are drawbacks/cons of using WAN:

- ✓ The initial setup cost of investment is very high.
- ✓ It is difficult to maintain the WAN network. You need skilled technicians and network administrators.
- ✓ There are more errors and issues because of the wide coverage and the use of different technologies.
- ✓ It requires more time to resolve issues because of the involvement of multiple wired and wireless technologies.
- ✓ Offers lower security compared to other types of networks.

What is Virtual Private Network?

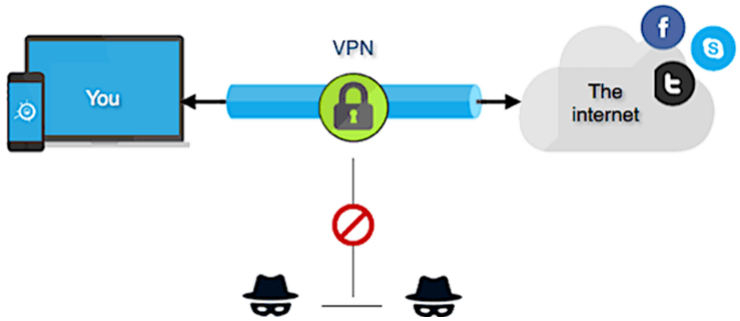

Figure 5.4 Virtual Private Network (VPN) Setup
https://networkencyclopedia.com/virtual-private-network-vpn/

A *Virtual Private Network*, or *VPN*, is an encrypted connection over the Internet from a device to a network. The encrypted connection helps ensure that sensitive data is safely transmitted. It prevents unauthorized people from eavesdropping on the traffic and

allows the user to conduct work remotely. VPN technology is widely used in corporate environments (Deeksha, 2020).

Characteristics of Virtual Private Network (VPN) (Deeksha, 2020)

1. **The Sign-Up and Payment Process** (Deeksha, 2020)

   The easier signing up and paying for a USA or Netherlands VPN is, the more satisfying the overall service will be. Nothing is worse than having problems with a service which is completely unrelated to the hardware and software. Encountering problems during the sign-up or payment phase is not only a headache, but an unnecessary frustration. One of the biggest issues during this phase is the payment processor. The more payment options a user has, the easier it will be to find one that works for them.

2. **Customer Support Availability** (Deeksha, 2020)

   The easiest factor to overlook is the quality and availability of customer support. Ideally, customer support is never needed by anyone, however this is not realistic. For many people, poor customer service is a deal-breaker. VPN services which place a high priority on customer support inherently place a high value on their customers. When comparing the availability, both the available contact times and contact methods should be considered.

3. **The Software** (Deeksha, 2020)

   For someone using a VPN on a daily basis, the software is the source of the primary interaction. When looking at the software provided by a VPN service, the user must consider both the initial setup and day-to-day use. Along with basic software functionality, there are a number of additional features which simplify the user's experience such as an IP checking tool or server selection tool.

4. **Available VPN Protocols** (Deeksha, 2020)

   For a VPN service to be easy-to-use, they must offer multiple protocols to their users. There are a variety of protocols available emphasizing factors such as privacy, security, and connection speed. It is important that the service provides a protocol compatible with the user's priorities, preferences and devices. The protocol dictates how the user's device and server interact, which means it will play a large role in the day-to-day satisfaction level of anyone using the service.

5. **Compatibility with All Devices** (Deeksha, 2020)

If a VPN is truly easy use, it will be compatible with all of the operating systems and devices the customer uses. There are a number of services which focus exclusively on a certain type of VPN – such as desktop computer VPNs. While these services do one thing particularly well, their customers must go elsewhere for their other devices. By choosing an easy-to-use VPN service, a customer will be able to easily integrate and use all of their devices from any location.

Advantages of VPN (Deeksha, 2020)

1. **Bypass Geo-Restrictions**

Geo-Restrictions are a method of content control where the content providers choose what kind of users get to access their platforms. This control is done based on the geographical location of that user. It is possible for these platforms to see your geographical area because of your IP address.

The way a VPN works is it hides your real IP address. So, deploying a VPN will, as a result, hide your real geographical address. Thus, letting you access various platforms without restrictions.

2. **Increased Online Privacy**

When you're not using a VPN, any website you visit can see your real IP address. To the people who might not realize why this is a big deal; an IP address can reveal a lot about you which can be used against you. An IP address can reveal what country you're in, your city, your ISP and even your zip code.

When you surf the internet, you leave a footprint. ISPs can track this and sell that data to advertisers for them to create an accurate customer profile. They then target you with ads according to this footprint. With a VPN, your real IP address is hidden, freeing you from all this danger.

3. **Free from Online Censorship**

In some countries, the government imposes a ban against certain online services and websites. The Great Firewall of China is one example; it prohibits access to several applications and websites throughout the country.

You can use a VPN that gives you a US based IP; meaning your fake IP will belong to the USA so China's prohibition rules won't apply to you and you can browse the web without any restrictions. This also applies to any restrictions your ISP imposes.

4. **Safer Torrenting**

When you are downloading Torrents, yes it is very convenient but it also poses some threats. The total number of people that download and upload a torrent is called a Swarm. All of these people can see your real IP address when you are on that torrent. Not only can it reveal everything about you, some ISPs and governments also prohibit downloading torrents.

If they catch you doing this, they have the right to report you to the authorities. Who wants all that unnecessary trouble? A VPN will protect you from all this hassle by simply hiding your real IP address.

5. **Network Scalability**

First off, network scalability is the measure of how easily a network can add or remove devices that are on that network. A private network is basically a network that assigns its private IPs to the devices on that network. This kind of network is preferred by a lot of companies. The budget required to create a private network is huge.

A VPN reduces this cost and the problem of scalability just by making use of the public network. Its use is even more significant for international connections.

Disadvantages of Virtual Private Network

1. **Illegal**

There are some countries that consider a private network, illegal like Uganda and Turkey. Some other countries are also working to ban the use of VPNs.

So, before you decide to buy a VPN service, make sure your country isn't one of the countries that want to ban VPN. Using it despite the ban might result in you having to pay a heavy fine.

2. Decreased Performance

When you use a VPN, what it does is it connects your network to a private server before letting you access a website. The problem with this is the private server might not respond right away to your request for a private network. What this means to you is that the website you are trying to access will take longer to load.

3. Monitoring by the Service

Yes, the whole point of getting a VPN is to avoid all the unnecessary monitoring and tracking online. But, even then some VPN services do monitor and track your data. It doesn't mean that their intentions are malicious all the time, but it is a possibility.

There is also a chance that the service will not tell you that they'll be monitoring you. This is the case with free VPNs or the ones that cost much less than other standard VPN services. So, that is another thing to keep in mind before paying for or downloading such a service.

4. Unable to Bypass 100%

The biggest reason why VPNs are so popular is because it allows access to services and websites that might have been inaccessible in that geographic area. It is easier with a VPN to access those websites but even VPNs cannot allow access to certain sites.

Hulu is an example of such a site. You can only access it if you live in the USA and a person from say, Nepal, cannot access it even with the use of a VPN. This is because of the agreement they have with content producers. They have to abide by it and ban it from certain areas.

5. Increased Cost

As mentioned before, some VPNs monitor your online activity. Some of these free services also do not work as intended, leaking your private internet information.

Other Types of Networks

Apart from above mentioned here, are some other important types of networks:

- WLAN (Wireless Local Area Network)
- Storage Area Network
- System Area Network
- Home Area Network
- POLAN- Passive Optical LAN
- Enterprise private network
- Campus Area Network

1. WLAN

WLAN (Wireless Local Area Network) helps you to link single or multiple devices using wireless communication within a limited area like home, school, or office building. It gives users an ability to move around within a local coverage area which may be connected to the network. Today most modern day's WLAN systems are based on IEEE 802.11 standards.

2. Storage-Area Network (SAN)

A Storage Area Network is a type of network which allows consolidated, block-level data storage. It is mainly used to make storage devices, like disk arrays, optical jukeboxes, and tape libraries.

3. System-Area Network

System Area Network is used for a local network. It offers high-speed connection in server-to-server and processor-to-processor applications. The computers connected on a SAN network operate as a single system at quite high speed.

4. Passive Optical Local Area Network

POLAN is a networking technology which helps you to integrate into structured cabling. It allows you to resolve the issues of supporting Ethernet protocols and network apps. POLAN allows you to use optical splitter which helps you to separate an optical signal from a single-mode optical fiber. It converts this single signal into multiple signals.

5. *Home Area Network (HAN):*

A Home Area Network is always built using two or more interconnected computers to form a local area network (LAN) within the home. For example, in the United States, about 15 million homes have more than one computer. This type of network helps computer owners to interconnect with multiple computers. This network allows sharing files, programs, printers, and other peripherals.

6. *Enterprise Private Network :*

Enterprise private network (EPN) networks are build and owned by businesses that want to securely connect numerous locations in order to share various computer resources.

7. *Campus Area Network (CAN):*

A Campus Area Network is made up of an interconnection of LANs within a specific geographical area. For example, a university campus can be linked with a variety of campus buildings to connect all the academic departments.

Lesson 3. Components of a Network

Computer network components include the key parts that are required to install a network. In computer network technology, there are numerous types of networks that vary from simple to complex level (Cloud, 2015).

These hardware components involve cable, Hub, Switch, NIC (network interface card), modem and router. Some of the components can be removed without effect subject to the type of network. For instance, in a wireless network, cables and hubs are not necessary (Cloud, 2015).

The essential computer network components necessary for a simple computer network, usually called LAN (local area network), are to share resources on that network. Sharing of equipment like printers, scanners, fax machines, copiers, storage drives, network connection and other resources is the key rationale of computer network (Cloud, 2015).

MAJOR COMPUTER NETWORK COMPONENTS

Computer network involves the following components and devices (some are optional) such as Network Interface Card (NIC), Hub, Switches, Router, Modem, Cables and connectors, Software and Servers.

Network Interface Card

Figure 5.5 A stand-alone NIC used in a Desktop Computer

Source: http://cloudcomputingnet.com/computer-network-components/

Network adapter or NIC is a device which allows a computer to communicate with another computer/network. By means of a unique hardware addresses (MAC address) programmed on the card chip, the data-link protocol uses these addresses to detect other systems on the network so that it can transfer data to the correct destination via the wired or wireless network (Deeksha, 2020).

There are two forms of network cards: wired and wireless. The wired NIC employs cables and connectors as a medium to transfer data, while in the wireless card, the connection is made using antenna that uses radio wave technology. All current laptop computers are integrated with wireless NIC in addition to the wired adapter. There is also a portable USB based adapter that can be plugged into the USB port of the computer (Deeksha, 2020).

Network Interface card, one of the key computer network components, comes with different speeds of 10Mbps, 100Mbps and 1000Mbps, etc. Current standard network cards are made with Gigabit (1000Mbps) connection speed which supports slower speeds such as 10Mbps and 100Mbps as well. However, the actual transmission speed is determined by the LAN speed (Deeksha, 2020).

Network Hub

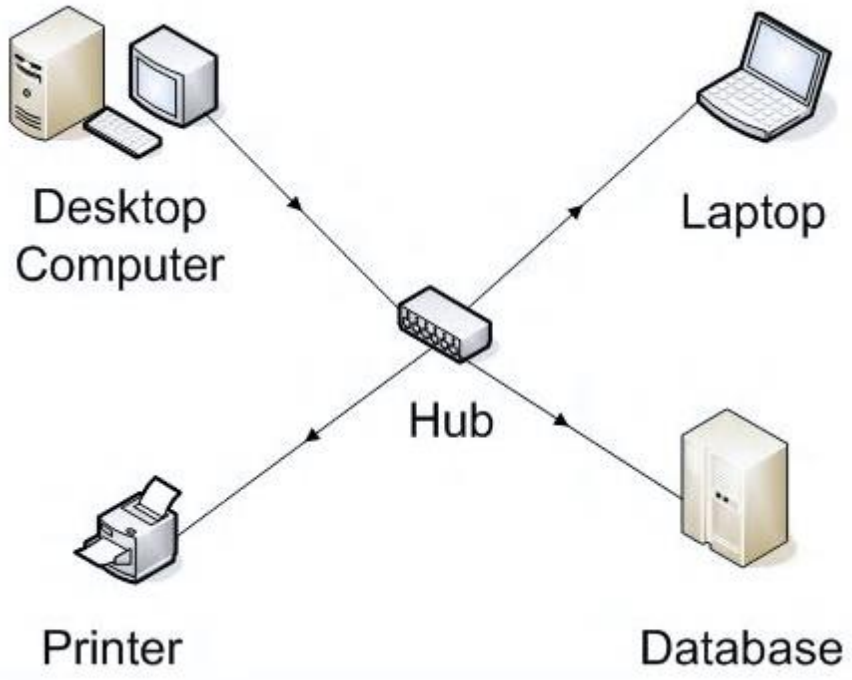

Figure 5.6 Hub Setup

Source: http://cloudcomputingnet.com/computer-network-components/

Hub is a network component that divides a network connection like a central dissemination center for multiple devices. When a computer request information from a network or a particular computer, it transmits the request to the hub through a cable. The hub receives the request and broadcasts it to the entire network. Each device in the network would then work out if the broadcast data is for them or just drop it if it is not. This consumes more bandwidth on the network and limits the amount of communication (Deeksha, 2020).

Currently Hubs are almost obsolete and superseded by more advanced network communication equipment such as Switches and Routers.

Network Switches

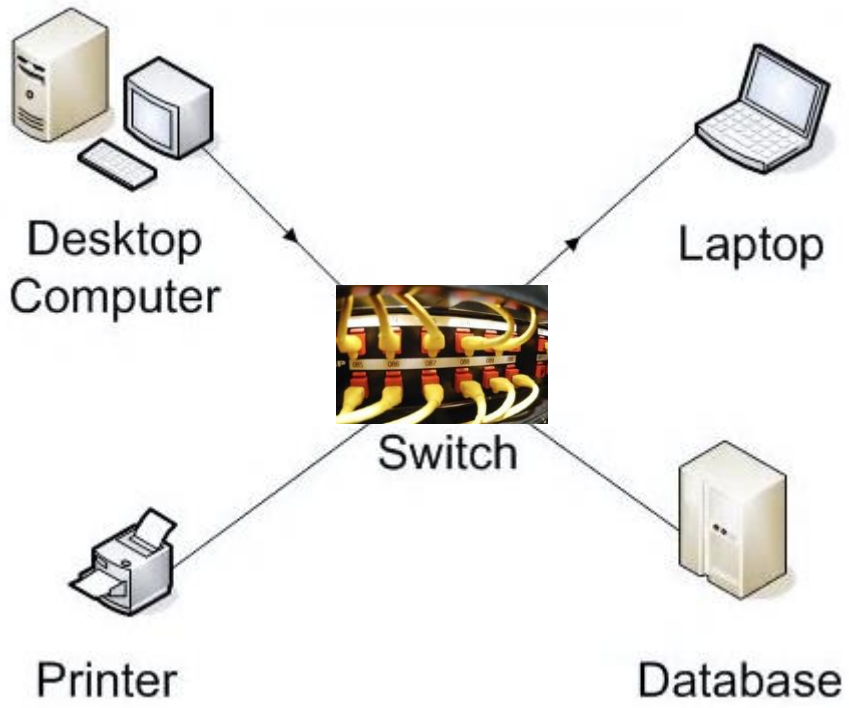

Figure 5.7 Switch Setup

Source: http://cloudcomputingnet.com/computer-network-components/

Switch is like a smart Hub made in with advanced features like the ability to form tables that keep updated about all the computer or device and on which switch port it is on. Using the physical device addresses in each incoming message, it delivers the message to the correct destination or port. Nevertheless, both switch and hub have similar features: Several RJ-45 ports, power supply and connection lights (Deeksha, 2020).

Switches do not broadcast the received message to entire network like the Hub but rather before transmitting, it analyses which system or port should the message be transmitted to. The

switch connects the source and destination directly which increases the speed of the network (Deeksha, 2020).

Switching technology aids to ease congestion on a network and is usually used for networks of ten or more devices.

Network Router

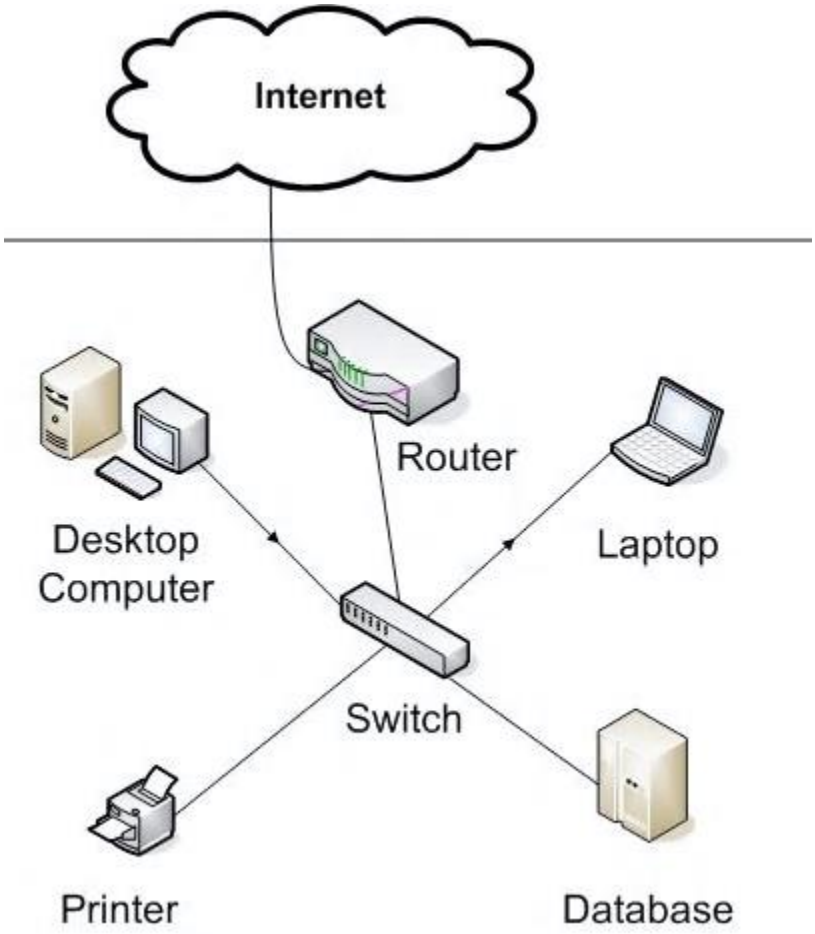

Figure 5.8 Router Setup

Source: http://cloudcomputingnet.com/computer-network-components/

A Router is the network component that is employed to connect a LAN to an internet connection. Routers are similarly employed to link one local area network (LAN) to another, often across long distances through commercial carriers (Deeksha, 2020).

Routers are actually smart switches in that they are conscious of other networks. They can dynamically update their routing information, discovering when a route to a network is down and finding if another route is accessible (Deeksha, 2020).

Generally, routers include a switch which means that it can be used as a switch as well. The router portion links the network to the Internet and the switch portion allows the computers to communicate with each other and to the internet (Deeksha, 2020).

The choice of wired or wireless routers is influenced by on the physical setting, speed and cost.

Network Modems

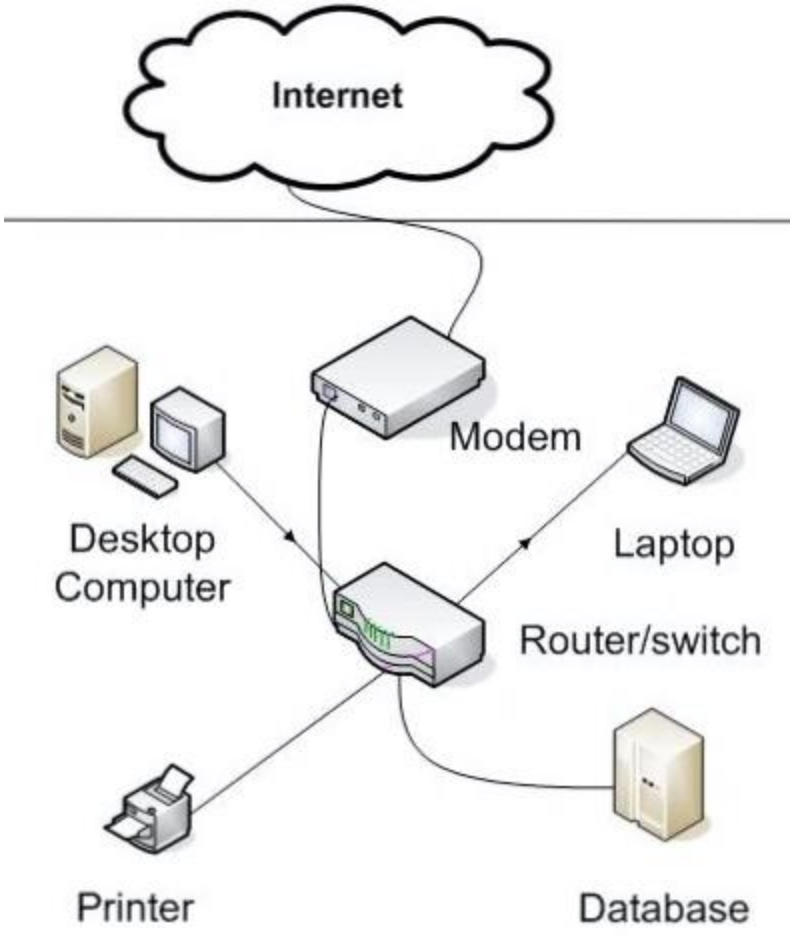

Figure 5.9 Modem Setup

Source: http://cloudcomputingnet.com/computer-network-components/

An analogue or digital modem allows connection to the internet over the existing telephone line. These types of modem are not usually integrated with a computer motherboard but come as separate devices which can be installed on the PCI slots found on motherboard or a standalone device. A modem is not required for LAN but needed for internet connection via dial-up and DSL (Cloud, 2015).

A broadband modem is utilized to yield a high-speed Internet connection provided by an ISP (Internet Service Provider). The high-speed connection can be DSL (Digital Subscriber Line) from a telephone company or cable (cable modem) from a cable television provider. Cellular modem is employed when the cellular data mobile network (3G and above) is used to access the internet, usually provided by cellular mobile operators (Cloud, 2015).

Fiber Broadband Modem is used to achieve even higher speed connection, in the Gigabits range to the internet via optical fiber network.

Network Cables and Connectors (Cloud, 2015)

The networking devices will not work well unless they are connected to each other which is done via different media. Cable is one transmission media which can transmit communication signals. The wired network topology uses special type of cable to link computers on a network. There are several transmission media types, some of which are listed below.

- **Twisted Pair Wire -** the most universal cable media is commonly named Ethernet cabling and it is categorized as Category 1, 2, 3, 4, 5, 5E, 6 and 7. The categories 5E, 6 and 7 are meant for high-speed that can transmit 1Gbps or more.
- **Coaxial Cable -** is similar to TV installation cable and it is costlier than twisted-pair cable, however, provides high data transmission speed.
- **Fiber-Optic Cable/Optical Fiber -** is a high-speed cable which transmits data via light beams (uses laser or LED light instead of electrical pulses) in glass bound fibers. Even though it offers higher data speed compared to the other cables, fiber optics is very costly making the purchase and installation at the governmental or large enterprise level.
- **Wireless Network Components -** has become prevalent in homes and small offices owing to the ease and low cost of setting up a network. The “media” in this case is the air, through which the wireless NICs transmit radio signals carrying data. The most widespread wireless protocol is acknowledged as WIFI.

Figure 5.10 Network Cables and Connectors

Source: http://cloudcomputingnet.com/computer-network-

Network Software (Cloud, 2015)

Software is the intelligence that instructs all the network components to operate together. The most prevalent network software employs what is known as the TCP/IP protocol suite or stack. The suite is structured from actual layers of software, each having its own function. Using the seven-layer OSI reference model - Physical, Data Link, Network, Transport, Session, Presentation and Application - as the starting point, the Internet model has four layers - Link, Internet, Transport and Application - combining several of the seven OSI layers into the other layers. All heterogeneous computer systems can communicate with each other using the same set of rules provided by the suite, irrespective of variation in hardware or operating systems.

Network Servers (Cloud, 2015)

Servers are digital banks for mass storage of data. Networks use servers to hold the data users may wish to access and to offer storage space. For instance, on the Internet, email providers utilize their servers to store the messages users send while web hosts use servers to hold all the data that a web page displays to visitors.

Lesson 4. E-Commerce

Electronic Commerce, commonly written as E-Commerce, is the trading in products or services using computer networks, such as the Internet. Electronic commerce draws on technologies such as mobile commerce, electronic funds transfer, supply chain management, Internet marketing, online transaction processing, electronic data interchange (EDI), inventory management systems, and automated data collection systems. Modern electronic commerce typically uses the World Wide Web for at least one part of the transaction's life cycle, although it may also use other technologies such as E-Mail (Kutz, 2016).

E-Commerce businesses may employ some of or all of the following:

- Online shopping websites for retail sales direct to consumers,
- Providing or participating in online marketplaces, which process third-party business-to-consumer or consumer-to-consumer sales,
- Business-to-business buying and selling,
- Gathering and using demographic data through Web contacts and social media,
- Business-to-business electronic data interchange,
- Marketing to prospective and established customers by E-Mail or fax (for example, with newsletters), and
- Engaging in pretail for launching new products and services.

Pretail (al referred to as pre-tail, or pre-commerce) is a sub-category of E-Commerce and online retail for introducing new products, services, and brands to market by pre-launching online, sometimes as reservations in limited quantity before release, realization, or commercial availability. Pretail includes pre-sale commerce, pre-order retailers, incubation marketplaces, and crowdfunding communities (Kutz, 2016).

E-Business

Electronic Business, or E-Business, is the application of information and communication technologies (ICT) in support of all the activities of business. Commerce constitutes the exchange of products and services between businesses, groups and individuals and can be seen as one of the essential activities of any business. Electronic commerce focuses on the use of ICT to enable the external activities and relationships of the business with individuals, groups, and other businesses or E-Business refers to business with help of Internet i.e. doing business with the help

Internet network. The term <E-Business> was coined by IBM's marketing and Internet team in 1996 (Kutz, 2016).

Comparing E-Commerce and E-Business we come to the subsequent conclusion: *"E-Business is a more general term than E-Commerce. However, in this module we will only use the term E-Commerce, because very business transaction finally is involved in selling or buying of products or services. And the term E-Commerce obviously is more widespread than the term E-Business."*

Digital Economy

Digital Economy refers to an economy that is (substantially) based on computing technologies. The digital economy is also sometimes called the Internet Economy, the New Economy, or Web Economy. Increasingly, the "digital economy" is intertwined with the traditional economy making a clear delineation harder (Kutz, 2016).

We will not use the term "digital economy" further on this module, because business is business be it traditional or digital. And boundaries are moving everyday due to technical development. However, we will repeatedly use the term "digital" or "digitalized" to indicate that subjects or activities are based on ICT.

PRELIMINARY DEFINITION (Kutz, 2016)

Some authors write extremely enthusiastically like this: E-Commerce enables the comprehensive digital execution of business processes between suppliers and their customers via global public and private networks. However, this definition rises some questions:

- What does "comprehensive" mean? Does it mean the total process? Is everything digitalized?
- What about transportation and delivery of real goods? Obviously here are some limits for digitalization, though sooner or later 3-D-printing may change a lot.
- Why should businesses be run electronically? Is enablement a value in itself? Or do we digitalize businesses because we can reduce costs, accelerate processes, and increase profit?

This definition, though given in many E-Commerce modules, is too much marketing-minded and not helpful to understand the advantages (and disadvantages) of "digitalized" business reasonably.

FINAL DEFINITION (Kutz, 2016)

To come to a final definition of E-Commerce, let us start with some constituent attributes of E-Commerce:

- Digitalization of Business:
  - This means a comprehensive usage of Information and Communication Technology (ICT) not only within a business organization (as it has been done during the last decades by traditional (internal) information systems), but now, through a more and more seamless linking and cooperation of information and communication systems of all involved business partners.
  - The comprehensive usage of ICT has been enabled by some technologies and technical standards, which have been accepted globally.

- Focus on Business Processes:
  - We support business processes, of course, as we did it for the last decades, but now the total processes, running through several organizations and crossing their boundaries, are supported.
  - We automate business process not longer only within organizations, as it was the traditional objective of ICT, but now the automation is related to the total process, running through all involved organizations, and not only to the sub-process within the own organization.
  - We increase the speed of business processes. Additional potentials can be realized with the coupling of processes between different organizations.
  - We increase the economic efficiency of business processes, again through coupling of business processes at the boundaries of the business partners.

- Usage of a Global Network
  - Internet plays a dominant role and has become a universal technical infrastructure. Thus, it builds a global virtual place where every organization and person being interested in making business can come together without geographical and time restrictions.

- Global networks all the exchange of information without any restrictions in time and independently from any geographical distances.
- We know (means assume) that the Internet is always up and running (24/7).

- New Potentials and Opportunities for Cooperation
  - More or less independent persons and/or organizations work together.
  - Business actors can come together whenever they want it or whenever there is a need.

These considerations lead to our final definition (Turban, et. al. 2015, p. 7):

> *"E-Commerce is the exchange of goods and services between (usually) independent organizations and/or persons supported by a comprehensive usage of powerful ICT systems and a globally standardized network infrastructure.*
>
> *For this purpose, the business partners to couple their business processes and their ICT systems. These systems have to work together temporarily and seamlessly and have to share, exchange and process data during the whole business process and across the boundaries of the cooperating organizations.*
>
> *Data security and data privacy as well as the compliance with laws and other policies and procedures have, of course, to be guaranteed.*

E-Commerce with the "5-C-Model"

Another approach to define and explain, what E-Commerce is, comes from the so-called 5-C-Model (Zwaas, 2014). It defines E-Commerce by five activity domains whose denominations start with the letter "C".

Commerce

- In the electronic marketplaces there is a matching customers and supplies, an establishing of the transaction terms, and the facilitation of exchange transactions.

- With the broad move to the Web-enable enterprise system with relatively uniform capabilities as compared to the legacy systems, a universal supply-chain linkage has been created.

Collaboration

- The Web is a vast nexus, or network, of relationships among firms and individuals.
- More or less formal collaborations are created or emerge on the Web to bring together individuals engaged in knowledge work in a manner that limits the constraints of space, time, national boundaries, and organizational affiliation.

Communication

- As an interactive medium, the Web has given rise to a multiplicity of media products.
- This universal medium has become a forum for self-expression (as in blogs) and self-presentation (as, for an example, in Polyvore: www.polyvore.com).
- The rapidly growing M-Commerce enables connectivity in context, with location-sensitive products and advertising.
- In the communication domain, the Web also serves as a distribution channel for digital products.

Connection

- Common software development platforms, many of them in the open-source domain, enable a wide spectrum of firms to avail themselves of the benefits of the already developed software, which is, moreover, compatible with that of their trading and collaborating partners.
- The Internet, as a network of networks that is easy to join and out of which it is relatively easy to carve out virtual private networks, is the universal telecommunications network, now widely expanding in the mobile domain.

Computation

- Internet infrastructure enables large-scale sharing of computational and storage resources, thus leading to the implementation of the decades-old idea of utility computing.

ADVANTAGES AND DISADVANTAGES OF E-COMMERCE (Kutz, 2016)

E-Commerce has a lot of advantages. But as we know it from every area for our life, there is “no free lunch”. Of course, E-Commerce has some disadvantages.

Table 5.2 Advantages of E-Commerce

ADVANTAGES	
…for the customer	*…for the provider*
▪ Flexible shopping hours (24/7) ▪ No waiting queues (if net is available and software appropriately designed) ▪ Shopping at home (we don’t have to leave our apartment, refuel our car or buy a subway ticket, look for a parking place, etc.) ▪ Individual needs can be covered (if customization is offered) ▪ Global offers, more competition, pressure on prices	▪ Better customer service can be offered ▪ Fast communication with customer ▪ New customer potential through global visibility ▪ No (traditional) intermediaries, who take away margins

Table 5.3 Disadvantages of E-Commerce

DISADVANTAGES	
…for the customer	*…for the provider*
▪ Security Risk: - Data theft (e.g. stealing account or credit card numbers) - Identify theft (acting under our name or user identity) - Abuse (e.g. third person orders goods with our identity, gets them delivered and we have to pay for it) ▪ Crime:	▪ Higher logistics cost (goods have to be sent to the customer’s location) ▪ Anonymity of customers (how to make targeted advertisements?

- Bogus firm (firm does not really exist) - Fraud (e.g. order is confirmed, invoice has to be paid, but goods are never delivered) ▪ Uncertain legal status (if something goes wrong, can we accuse the provider?)	

Lesson 5. Internet or the Internetwork (Kutz, 2016)

What is the Internet?

Before we learn about how to use the internet, it is good to define what exactly the internet is. The Internet is a worldwide network of computers that allows information from educational facilities, companies, government, and individuals to be shared, or "networked". Over 100 countries are linked to one another and can share their news and opinions across the globe (Kutz, 2016).

Brief History

The Internet (portmanteau of interconnected network) is the global system of interconnected computer networks that use the Internet protocol suite (TCP/IP) to link devices worldwide. It is a network of networks that consists of private, public, academic, business, and government networks of local to global scope, linked by a broad array of electronic, wireless, and optical networking technologies. The Internet carries a vast range of information resources and services, such as the inter-linked hypertext documents and applications of the World Wide Web (WWW), electronic mail, telephony, and file sharing (Kutz, 2016).

The origins of the Internet date back to research commissioned by the federal government of the United States in the 1960s to build robust, fault-tolerant communication with computer networks. The primary precursor network, the ARPANET, initially served as a backbone for interconnection of regional academic and military networks in the 1980s. The funding of the National Science Foundation Network as a new backbone in the 1980s, as well as private funding for other commercial extensions, led to worldwide participation in the development of new networking technologies, and the merger of many networks. The linking of commercial networks and enterprises by the early 1990s marked the beginning of the transition to the modern Internet and generated a sustained exponential growth as generations of institutional, personal, and mobile computers were connected to the network. Although the Internet was widely used by academia since the 1980s, commercialization incorporated its services and technologies into virtually every aspect of modern life (Kutz, 2016).

Most traditional communication media, including telephony, radio, television, paper mail and newspapers are reshaped, redefined, or even bypassed by the Internet, giving birth to new services such as email, Internet telephony, Internet television, online music, digital newspapers, and video streaming websites. Newspaper, book, and other print publishing are adapting to website technology, or are reshaped into blogging, web feeds and online news aggregators. The Internet has enabled and accelerated new forms of personal interactions through instant messaging, Internet forums, and social networking. Online shopping has grown exponentially both for major retailers and small businesses and entrepreneurs, as it enables firms to extend their "brick and mortar" presence to serve a larger market or even sell goods and services entirely online. Business-to-business and financial services on the Internet affect supply chains across entire industries (Kutz, 2016).

The Internet has no single centralized governance in either technological implementation or policies for access and usage; each constituent network sets its own policies. The overreaching definitions of the two principal name spaces in the Internet, the Internet Protocol address (IP address) space and the Domain Name System (DNS), are directed by a maintainer organization, the Internet Corporation for Assigned Names and Numbers (ICANN). The technical underpinning and standardization of the core protocols is an activity of the Internet Engineering Task Force (IETF), a non-profit organization of loosely affiliated international participants that anyone may associate with by contributing technical expertise. In November 2006, the Internet was included on USA Today's list of New Seven Wonders (Princetone, 2020).

What is Internet Explorer?

Internet Explorer is a program that can be used to search the World Wide Web for the information it contains. It is called a 'web browser' because it is used to browse the World Wide Web. It is the most commonly used web browser as it is forming a part of the Microsoft operating system that most computers use. For this reason, in this module, we will be focusing on the features of this web browser (Princetone, 2020).

Launching Internet Explorer

To begin using Internet Explorer, you need to double click with the left mouse button on the Internet Explorer icon which may be found on your Desktop (the initial screen when your computer first starts up).

If you do not have an icon on your desktop, click on the Start menu in the bottom left-hand corner of your screen, go to Programs and select Internet Explorer from the list of options in the Programs menu.

If you would like to create a Shortcut on your desktop for future use, right click the mouse on the Internet Explorer listing. This will bring up a list of options from which you can choose to 'Send to Desktop (create shortcut)'. If you look at your desktop now, an Icon for Internet Explorer will have appeared (Princetone, 2020).

Internet Explorer Window (Princetone, 2020)

Once you have double-clicked on the Internet Explorer shortcut on your desktop to launch the browser, the main screen, or 'window', of the program will appear. The Main Window has many parts such as the Title Bar, Menu Bar, Toolbar, Address Bar, Link Bar, Main Browser Window as well as the Status Bar. These are shown in the diagram below and explained in the table.

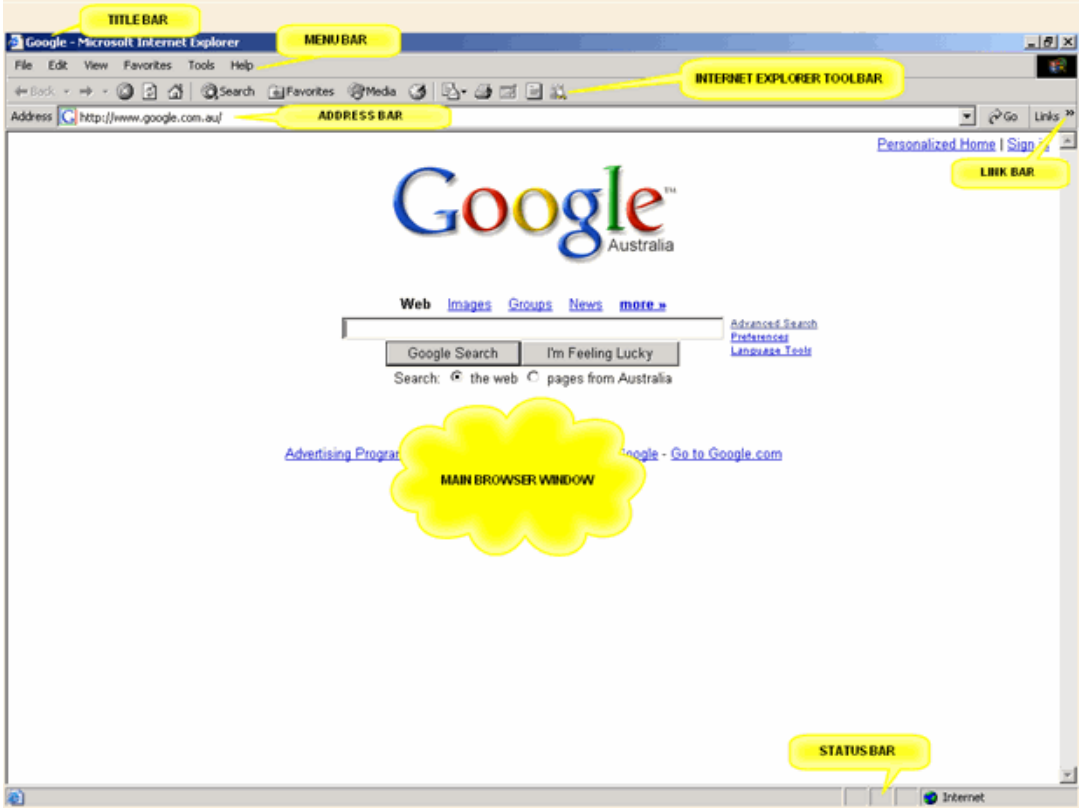

Figure 5.11 Google Interface using Internet Explorer

Source: https://www.thecomputerschool.net/modules/internet/internet_module.html

Table 5.4 Parts of Internet Explorer

Title Bar	▪ Shows the Title of the Current Page. In the example given, the title of the page is Google.
Menu Bar	▪ Consists of many different Menus which control all the options, functions and commands of Internet Explorer.
Internet Explorer Toolbar	▪ Shows the most frequent commands and all browsing functions.
Address Bar	▪ Shows the address, or URL (Universal Resource Locator), of the page that is currently open. You can type any webpage address into this bar and press Enter on the keyboard (or GO to the right of the address bar) to open it.
Link Bar	▪ Contains the links to the main Microsoft home site. You can customize links by dragging (holding the mouse button down while

	you move across the screen) the internet icon in the address bar from the Address Bar to the Link Bar.
Main Browser Window	▪ Displays all the information and contents of the website currently open. The window may have Scroll Bars located on the right side and on the bottom, which allow you to view the parts of the pages that are too large to fit on the screen.
Status Bar	▪ Shows the progress of the browser and information like download file status, document security, and link addresses.

Browsing the Web (Princetone, 2020)

- Browsing is a term that typically refers to viewing webpages.
- Web Browsers are programs used to view webpages. There are many Web Browsers available to view the Internet such as Netscape Navigator, Mozilla Firefox and, the one that we are becoming more familiar with in this module, Internet Explorer.

Points To Remember (Princetone, 2020)

- Each webpage has its own unique address.
- Web Addresses may be case sensitive, so always type carefully and exactly as it was given to you.
- There are never any blank spaces in a web address.
- If you want to change the website you are visiting, delete the current address and type the new Web Address in the Address Bar and press Enter.
- If the Web Address begins with http://, you can leave this part off and type the remaining part of the address as Internet Explorer automatically adds it.
- A list of previously entered Web Addresses can be viewed by clicking the drop-down list on the right-hand side of the address field
- Whenever you see a hand symbol replace the mouse pointer, you are hovering over what is known as a hyperlink. A hyperlink is an area of a webpage, usually blue and underlined if it is text that contains a web address that you will automatically follow if you click on it. Sometimes it will open the new link in a new window, but other times it will replace the one that you are currently viewing. A way to ensure that it will open a new window is to right click the mouse button when you are on the hyperlink and select Open in New Window.

Most Common Search Engines (Princetone, 2020)

Here is a list of some of the most common Search Engines:

- http://www.google.com
- www.youtube.com
- http://www.yahoo.com

To use one of these search engines, type the address into the Address Bar of Internet Explorer and then press Enter on the keyboard. Type a word or phrase relating to what you would like information on in the search field and click on the Search button located beside it. The Search Engine will then give you a list of results that match your search.

Lesson 6. IPs, Protocols, Browsers, URL

In short, an Intrusion Prevention System (IPS), also known as intrusion detection prevention system (IDPS), is a technology that keeps an eye on a network for any malicious activities attempting to exploit a known vulnerability (computerhope.com, 2020).

An Intrusion Prevention System's main function is to identify any suspicious activity and either detect and allow (IDS) or prevent (IPS) the threat. The attempt is logged and reported to the network managers or Security Operations Center (SOC) staff.

Why should Intrusion Prevention Systems be used? (Princetone, 2020)

IPS technologies can detect or prevent network security attacks such as brute force attacks, Denial of Service (DoS) attacks and vulnerability exploits. A vulnerability is a weakness in a software system and an exploit is an attack that leverages that vulnerability to gain control of a system. When an exploit is announced, there is often a window of opportunity for attackers to exploit that vulnerability before the security patch is applied. An Intrusion Prevention System can be used in these cases to quickly block these attacks.

Because IPS technologies watch packet flows, they can also be used to enforce the use of secure protocols and deny the use of insecure protocols such as earlier versions of SSL or protocols using weak ciphers.

How do Intrusion Prevention Systems work? (Princetone, 2020)

IPS technologies have access to packets where they are deployed, either as Network intrusion detection systems (NIDS), or as Host intrusion detection systems (HIDS). Network IPS has a larger view of the entire network and can either deployed inline in the network or offline to the network as a passive sensor that receives packets from a network TAP or SPAN port.

The detection method employed may be signature or anomaly-based. Predefined signatures are patterns of well-known network attacks. The IPS compares packet flows with the signature to see if there is a pattern match. Anomaly-based intrusion detection systems uses heuristics to identify threats, for instance comparing a sample of traffic against a known baseline.

What's the difference between IDS and IPS? (Princetone, 2020)

Early implementations of the technology were deployed in detect mode on dedicated security appliances. As the technology has matured and moved into integrated Next Generation Firewall or UTM devices, the default action is set to prevent the malicious traffic.

In some cases, the decision to detect and accept or prevent the traffic is based upon confidence in the specific IPS protection. When there is lower confidence in an IPS protection, then there is a higher likelihood of false positives. A false positive is when the IDS identifies an activity as an attack, but the activity is acceptable behavior. For this reason, many IPS technologies also could capture packet sequences from the attack event. These can then be analyzed to determine if there was an actual threat and to further improve the IPS protection.

Communications Protocols (Princetone, 2020)

What are Protocols?

Simply having a wired or wireless physical connection between end devices is not enough to enable communication. For communication to occur, devices must know "how" to communicate. Communication, whether by face-to-face or over a network, is governed by rules called protocols. These protocols are specific to the type of communication method occurring.

For example, consider two people communicating face-to-face. Prior to communicating, they must agree on how to communicate. If the communication is using voice, they must first agree on the language. Next, when they have a message to share, they must be able to format that message in a way that is understandable. For example, if someone uses the English language, but poor sentence structure, the message can easily be misunderstood.

Similarly, network protocols specify many features of network communication, as shown in the figure.

Network Protocols (Princetone, 2020)

Network protocols provide the means for computers to communicate on networks. Network protocols dictate the message encoding, formatting, encapsulation, size, timing, and delivery options. Networking protocols define a common format and set of rules for exchanging messages between devices. Some common networking protocols are Hypertext Transfer Protocol (HTTP), Transmission Control Protocol (TCP), and Internet Protocol (IP). As a cybersecurity analyst, you must be very familiar with the structure of protocol data and how the protocols function in network communications.

Today, the TCP/IP protocol suite includes many protocols and continues to evolve to support new services.TCP/IP is the protocol suite used by the internet and the networks of today. TCP/IP has two important aspects for vendors and manufacturers:

- *Open Standard Protocol Suite*. This means it is freely available to the public and can be used by any vendor on their hardware or in their software.
- *Standards-Based Protocol Suite*. This means it has been endorsed by the networking industry and approved by a standards organization. This ensures that products from different manufacturers can interoperate successfully.

Message Formatting and Encapsulation (Princetone, 2020)

When a message is sent from source to destination, it must use a specific format or structure. Message formats depend on the type of message and the channel that is used to deliver the message.

Message Timing (Princetone, 2020)

Message timing is also very important in network communications. Message timing includes the following:

- **Flow Control.** This is the process of managing the rate of data transmission. Flow control defines how much information can be sent and the speed at which it can be delivered. For example, if one person speaks too quickly, it may be difficult for the receiver to hear and understand the message. In network communication, there are network protocols used by the source and destination devices to negotiate and manage the flow of information.
- **Response Timeout.** If a person asks a question and does not hear a response within an acceptable amount of time, the person assumes that no answer is coming and reacts accordingly. The person may repeat the question or instead, may go on with the conversation. Hosts on the network use network protocols that specify how long to wait for responses and what action to take if a response timeout occurs.
- **Access Method.** This determines when someone can send a message. Click Play in the figure to see an animation of two people talking at the same time, then a “collision of information” occurs, and it is necessary for the two to back off and start again. Likewise, when a device wants to transmit on a wireless LAN, it is necessary for the WLAN network interface card (NIC) to determine whether the wireless medium is available.

The Benefits of Using a Layered Model (Princetone, 2020)

You cannot actually watch real packets travel across a real network the way you can watch the components of a car being put together on an assembly line. So, it helps to have a way of thinking about a network so that you can imagine what is happening. A model is useful in these situations.

Complex concepts such as how a network operates can be difficult to explain and understand. For this reason, a layered model is used to modularize the operations of a network into manageable layers.

These are the benefits of using a layered model to describe network protocols and operations:

- Assisting in protocol design because protocols that operate at a specific layer have defined information that they act upon and a defined interface to the layers above and below
- Fostering competition because products from different vendors can work together
- Preventing technology or capability changes in one layer from affecting other layers above and below
- Providing a common language to describe networking functions and capabilities

Browsers (Princetone, 2020)

An internet browser, also known as a web browser or simply a browser, is a software program that lets you view web pages on your computer. You can think of your browser as your gateway to the internet. If you want to enter online sweepstakes, for example, you must first open a giveaway's website in your internet browser.

The main purpose of an internet browser is to translate the code that computers use to create websites into the text, graphics, and other features of the web pages that we're used to seeing today. Web browsers have evolved significantly since they were first introduced in 1990.

The First Web Browser (Princetone, 2020)

The first web browser was called Worldwide Web and later changed its name to Nexus. Created by Sir Tim Berners-Lee, it was released in 1990 and gave people a basic way to view web pages. However, it was a long way from the immersive online experience we have today.

Without browsers, the internet as we know it today would not be possible. The internet was text-based, bland, and couldn't be used without technical know-how. This limited the number of people who had the ability and the interest to use the internet.

In the early 90s, the audience was so small and so niche that most companies didn't have an online presence and maga shopping sites like eBay and Amazon would never have had a chance of drawing a decent crowd. That started to change with the introduction of Mosaic in 1992.

The Mosaic browser helped make the internet ubiquitous. The graphical interface made navigating the web easy to understand and the ability to display graphics on websites made web pages more interesting. Best of all, people no longer needed to be programmers to enjoy themselves online.

With more people spending more time on the internet, companies were quick to follow. E-commerce, online sweepstakes, social media, and many other things we take for granted today would be impossible without internet browsers.

Modern Internet Browsers (Princetone, 2020)

Internet browsers have developed into powerful tools that let you safely and quickly access your favorite websites. Modern internet browsers have many helpful features. Tabbed browsing, for example, helps you open up many web pages in individual tabs, instead of needing a resource-intensive separate window for each page. This is very helpful if you want to enter sweepstakes faster. Modern browsers offer security features, embedded video, nearly instant loading times, and many other things that make the net such a fun place to spend time.

Which Internet Browsers Are Available Today? (Princetone, 2020)

Most browsers are available for free download. The six most popular internet browsers today include:

- Mozilla Firefox (Download Firefox)
- Google Chrome (Download Chrome)
- Microsoft Edge (formerly Internet Explorer)
- Apple Safari (Download Safari)
- Opera (Download Opera)

Each of these internet browsers has advantages and disadvantages. Read their websites, then try out the ones that appeal to you the most. Due to the way they are coded, some websites display better in one internet browser or another. For this reason, it's a good idea to have at least two browsers available on your computer, especially if you want to enter sweepstakes. If an entry

form isn't displaying in your primary browser, it might work flawlessly on your secondary internet browser.

Many internet browsers support plug-ins, which are helper programs that you can download to customize your browsing experience. Plug-ins can help you fill out forms automatically, check your grammar and spelling, mute unwanted sounds in your browser, send you reminders, and much more.

Plug-ins can help you work or play online, but not every browser allows them. Firefox and Chrome each allow plug-ins, for example, but not all are available for each system.

The best way to find out which browser is right for you is to download them and play around with them. Which extra features are important to you? Do websites load quickly and reliably? Is your browser using too many resources and slowing down other programs you're using?

How Internet Browsers Work: (Princetone, 2020)

Here's a very quick overview of how browsers work:

- You type a website's URL into your browser's address bar; "http://www.thebalance.com" is an example of a URL.
- The browser locates and requests that page's information from a web server.
- The browser receives a file in a computer code like HTML or JavaScript, which includes instructions about how to display the information on that page.
- The browser interprets that file and displays the page for you to read and interact with. And it does all of this in just a few seconds, usually.

What is URL? (computerhope.com, 2020)

Also known as an internet address or web address, a URL (Uniform Resource Locator) is a form of URI and standardized naming convention for addressing documents accessible over the Internet and Intranet. An example of a URL is https://www.computerhope.com, which is the URL for the Computer Hope website.

Overview of a URL (computerhope.com, 2020)

Below is additional information about each of the sections of the http URL:

http:// or https:// (computerhope.com, 2020)

The "http" stands for Hypertext Transfer Protocol. It lets the browser to know which protocol it is going to use to access the information specified in the domain. An "https" protocol is short for "Hypertext Transfer Protocol Secure" and indicates that information transmitted over HTTP is encrypted and secure. After the http or https is the colon (:) and two forward slashes (//) that separate the protocol from the remainder of the URL.

Www. (computerhope.com, 2020)

Next, "www" stands for World Wide Web and is used to distinguish the content. This portion of the URL is not required and many times can be left out. For example, typing "http://computerhope.com" would still get you to the Computer Hope website. This portion of the address can also be substituted for an important subpage known as a subdomain.

computerhope.com (computerhope.com, 2020)

Next, "computerhope.com" is the domain name for the website. The last portion of the domain is known as the domain suffix, or TLD. It is used to identify the type or location of the website. For example, ".com" is short for commercial, ".org" is short for an organization, and ".co.uk" is the United Kingdom. There are several domain suffixes available. To get a domain, you would register the name through a domain registrar.

url.htm (computerhope.com, 2020)

Finally, url.htm is the actual web page on the domain you're viewing. The trailing .htm is the file extension of the web page that indicates the file is an HTML file. Other common file extensions on the Internet include .html, .php, .asp, .cgi, .xml, .jpg, and .gif. Each of these file extensions performs a different function, like all the different types of files on your computer.

Where is the URL located? (computerhope.com, 2020)

A URL is located at the top of the browser window in the address bar or omnibox depending on your browser window. On desktop computers and laptop, unless your browser is being displayed in full screen the URL is always visible. In most smartphone and tablet browsers, the address bar containing the URL will disappear as you scroll down and only show the domain when visible. When the address bar is not visible, scroll up the page. If only the domain is shown, tapping on the address bar shows the full address.

How to open an URL (computerhope.com, 2020)

You can open a URL by clicking a hyperlink. For example, if you click "hyperlink" in this paragraph, it opens a page describing hyperlinks.

If a URL is in printed material, (e.g., e-mail or magazine) where it's not a hyperlink, you can open the page by typing the URL in the browser address bar. If the URL is in an e-mail, it can also be copied and pasted into the address bar.

What characters are not allowed in a URL? (computerhope.com, 2020)

Most people realize that a space is not allowed in a URL. However, it is also important to realize, as documented in RFC 1738, the URL string can only contain alphanumeric characters and the !$-_+*'(), characters. Any other characters that are needed in the URL must be encoded.

Understanding more complex URLs and parameters (computerhope.com, 2020)

When a URL points to a script that performs additional functions, additional information (parameters) is added to the end of the URL. For example, a search engine URL pointing to a search results page includes a parameter with the search query words.

After the script file name is a ? (Question mark). The question mark in a URL separates the URL from all the parameters or variables being sent to the script. In the example above, the parameter being sent is q=example%20search. The "q" is a variable name, and the "example%20search" is the value being sent to that variable. Because no spaces are allowed in a URL, the space is encoded as %20. In many scripts, a + (plus) is also used to represent a space.

Is an IP address the same as a URL or web address?

No. An IP address is a unique number that's assigned to each device on a network. On the World Wide Web, a domain name is assigned a unique IP address. When typed (e.g., computerhope.com), DNS translates the domain name into an IP address that routers use to find the web server. A domain name is used instead of an IP address because it's easier for humans to remember. For example, it's easier to remember "computerhope.com" than it is to remember an IP address like "216.58.216.164" (computerhope.com, 2020)

Lesson 7. ISPs (Internet Service Providers)

Short for Internet Service Provider, it refers to a company that provides Internet services, including personal and business access to the Internet. For a monthly fee, the service provider usually provides a software package, username, password and access phone number. Equipped with a modem, you can then log on to the Internet and browse the World Wide Web and USENET and send and receive e-mail. For broadband access you typically receive the broadband modem hardware or pay a monthly fee for this equipment that is added to your ISP account billing (Beal, 2018).

In addition to serving individuals, ISPs also serve large companies, providing a direct connection from the company's networks to the Internet. ISPs themselves are connected to one another through Network Access Points (NAPs). ISPs may also be called IAPs (Internet Access Providers).

What is ISPs? (Beal, 2018)

ISP is short for Internet Service Provider, a company in your area that supplies you with access to the Internet and related services through several types of connection. There is a great number of ISPs throughout the world that offer various types of plans and fees, and the most well-known is AT&T, Verizon, Comcast Xfinity, etc. By the way, it is an ISP who provides you with an IP address - an identifier used while surfing the Internet.

When you make a request to open a web page or download a file through your connected to the Internet device, a whole chain of events occurs. First, the web browser you are using (e.g., Google Chrome, Mozilla Firefox, Internet Explorer, Safari, etc.) uses DNS servers to translate the alphabetic address of your request into its IP address. Then this IP address is sent from your router to your ISP, which in turn forwards it to the ISP of your request destination, and only then, in reverse order, the data you are searching for travels back to your device. However, it takes just a few seconds to perform this complicated process.

Types of ISPs (Beal, 2018)

In the 1990s, there were three types of ISPs: dial-up services, high-speed Internet (also referred to as "broadband") offered by cable companies, and DSL (Digital Line Subscribers) offered by phone companies. By 2013, dial-up services were rare (even though they were cheap), because they were very slow…and the other ISP options were typically readily available and much, much faster.

DSL and Cable. (Beal, 2018)

Two of the leading DSL ISPs have been Verizon and AT&T. But in the last few years (from 2013), DSL has been on the decline, while cable-based ISPs, like Comcast and Time Warner, have been growing. Why the change? It's because the phone companies have been getting more into the lucrative smartphone business, and selling annual contracts for cellular service along with smartphone Internet capabilities. That's left a lot of the broadband business for the cable companies.

Fiber Internet: On its way to you? (Beal, 2018)

With DSL dropping out of the picture, there's room for new technology and it's already here in some areas: its called fiber, or fiber optical, broadband. Supposedly, fiber is hundreds of times FASTER than cable or DSL. That's especially exciting news (if it's true and available) to companies, and gamers and households with a lot of simultaneous wireless usage going on.

Verizon (yes, they are downplaying DSL) now offers FiOS in select areas (put an "f" before "eye" and the "os"-sound in "most"). FiOS stands for fiber optic services, and it claims to have superfast Internet connection speeds.

And for all of us not in the Kansas area, Google launched Google Fiber in 2013, which offers incredibly ultra-fast Internet speed. Other companies (and communities) are teaming up to bring the next generation of broadband to you.

Lesson 8. Ethics or Netiquette (Beal, 2018)

In addition to formal policies, regulations, and/or laws which may govern use of computers and networks, the internet user community observes informal standards of conduct, often called "netiquette". These standards have evolved from users' experiences and their desire to foster an environment that makes using the internet a positive, productive, experience. Netiquette "no-no's" include:

- flaming (sending highly insulting or provoking email);
- asking questions before reading posted FAQ's (Frequently Asked Question lists);
- responding to entire mailing lists instead of to one person;
- sending email with rampant spelling or grammar errors;
- Forwarding hoaxes or chain mail.

Spamming, that is, causing the widespread distribution of unsolicited email, is not only annoying it also may cause excessive strain on computing facilities. Therefore it is prohibited by the University of California Electronic Communications Policy section on Appropriate Use: "III.D.7. Interference (link is external)".

The recommended response to netiquette infringements is to attempt to work directly with the perpetrator. If that is not successful, then contact the postmaster of the system which the person used.

Netiquette (Beal, 2018)

Netiquette is a combination of the words network and etiquette and is defined as a set of rules for acceptable online behavior. Similarly, online ethics focuses on the acceptable use of online resources in an online social environment.

Both phrases are frequently interchanged and are often combined with the concept of a ’netizen’ which itself is a contraction of the words internet and citizen and refers to both a person who uses the internet to participate in society, and an individual who has accepted the responsibility of using the internet in productive and socially responsible ways.

What Does Good Web Etiquette Look Like? (Beal, 2018)

Underlying this overall concept of socially responsible internet use are a few core pillars, though the details underneath each pillar are still subject to debate.

For Society:

- Recognizing that the internet is an extension of society. The internet isn't a new world in which anything goes, but rather, a new dimension of the world around us.
- Applying the same standards online as we do in public. In simple terms, this means that the values society has in place against hate speech and bigotry, child exploitation, and child pornography, copyright violations and other forms of theft, remain intact. Values around courtesy, kindness, openness, and treating others with the same respect we wish to receive should also be adhered to.
- Refusing to empower abuse and harassment while online. Accepting that the laws which are currently in place to protect the rights and dignity of citizens apply online and that where needed, laws are updated to reflect these rights in the extended environment. Theft, harassment, and bullying while online is still theft, harassment, and bullying, period.
- Acknowledging cultural differences. Even when national boundaries no longer apply, cultural respect and tolerance should remain. This requires finding a way to accept that the social values and norms of some netizens will not be the social values and norms of all netizens.

For Businesses:

For companies, being a good netizen, applying online ethics, and using netiquette include:

- Respecting rights of privacy for offline employees. Information possessed by citizens in their offline interactions should be respected.
- Maintaining transparency in information policies. By taking action so that consumers can easily and quickly understand how that company is using their information and protecting them from harm, companies can provide users with a clear means of ownership and self-determination as to what is, and isn't shared about them, which strengthens the consumer relationship.

Most internet users automatically apply the same responsible respectful behavior online as they do in every other environment and by nature apply netiquette an online ethics and are good netizens. The minority that fails to apply societal values in some or any environment- including the internet- are quickly identified as exceptions to be dealt with on a social, or criminal level. When you choose to partner with technology companies, especially for something as important

as internet security, it’s imperative you ensure that the partner shares your understanding of what it means to act ethically online.

The Core Rules of Netiquette by Virginia Shea

1. Remember the Human
2. Adhere to the same standards of behavior online that you follow in real life
3. Know where you are in cyberspace
4. Respect other people's time and bandwidth
5. Make yourself look good online
6. Share expert knowledge
7. Help keep flame wars under control
8. Respect other people's privacy
9. Don't abuse your power
10. Be forgiving of other people's mistakes

Lesson 9. Issues and Problems (Beal, 2018)

7 Common Network Issues and How to Resolve Them Fast

Networks are networks. Despite best efforts to keep things smooth all the time every day, things happen. Here's a look at some common network issues, some tips for quickly resolving them, and even better, how to prevent them from occurring again.

1. Duplicate IP Addresses

   When two devices attempt to share a single IP, you see the dreaded "Address Already in Use" Kill — with no ability to access the network.

   The Quick Fix: The blame for this often rests with your router's default DHCP configuration. DHCP is probably trying to assign your new device an address at the beginning of your subnet, and another device may already occupy these low-numbered addresses with static IPs. If you've just introduced a new device or server to your network, it may have its own DHCP server. Simply disable the DHCP server on that device to restore sanity to your network.

   The Preventive Measure: You can take one simple step to avoid IP conflicts by modifying your router's configuration to begin assigning DHCP addresses near the top end of your subnet, leaving the lower addresses available for devices that require static IPs.

2. IP Address Exhaustion

   To troubleshoot this issue, use the ipconfig command. If the workstation has assigned itself an IP address that begins with 169.x.x.x, it means that no IP address was available from the DHCP server.

   The Quick Fix: Some users on cable internet might not have a local router, in which case IP addresses are assigned on a limited basis directly from your ISP. You have probably run out of allowed IP addresses from your ISP. The solution to this is to purchase either a standalone router or Wi-Fi access point with an integrated router. This creates your own local pool of internal addresses, ensuring you won't run out. If you already have a local router with DHCP, the default address pool might be too small for your network. By accessing the DHCP settings on the router, you can adjust the size of the address pool to meet your network's needs.

The Preventive Measure: It's important that any internet-connected network have a local router in operation with NAT and DHCP, both for security reasons and to prevent IP address exhaustion. The router needs to be the only device connected to the modem, with all other devices connecting through the router.

3. **DNS Problems**

Errors such as The Network Path Cannot Be Found, IP Address Could Not Be Found, or DNS Name Does Not Exist, can usually be traced to a DNS configuration issue. The command line utility nslookup can be used to quickly show a workstation's DNS settings.

The Quick Fix: Workstations and other network devices can be configured to use their own DNS servers, ignoring the server assigned by DHCP. Checking the 'Internet Protocol Version 4 (TCP/IP)' settings for your adapter will show if an incorrect DNS server is specified, so just select "Obtain DNS server address automatically" instead.

The Prevention Measure: Your local router might be configured to operate as a DNS Server, creating a DNS pass-through to your ISPs servers. On busy networks, this may overload the capabilities of the router. Change your network's DHCP settings to directly access your DNS servers.

4. **Single Workstation Unable to Connect to the Network**

If only a single workstation is displaying the "No internet" message when opening a web browser, we can usually assume that the rest of the network is healthy and turn our attention to any hardware and software that is particular to this system.

The Quick Fix: To resolve this network issue, start by eliminating the obvious communication barriers such as a bad cable, poor Wi-Fi signal, failing network card or incorrect drivers. Ensure that the workstation's network adapter is configured with the correct IP, subnet, and DNS servers. If that doesn't solve the problem, check any firewall software on the device to ensure that necessary ports are open to the external network. Common ports include 80 and 443 for web traffic, plus 25, 587, 465, 110, and 995 for email.

The Preventive Measure: It's usually best to leave all workstation TCP/IP settings to "Automatically assigned." Use a DHCP server to hand out a uniform configuration to all devices on the network. If a static IP is needed on a particular workstation or server, most DHCP servers allow the ability to create static IP mappings.

5. **Unable to Connect to Local File or Printer Shares**

Sharing problems are among the most difficult network problems to solve, due to the number of components that need to be configured properly. Most commonly, sharing problems arise due to conflicts between mixed security environments. Even different versions of the same operating system sometimes use slightly different security models, which can make interconnection of workstations difficult.

The Quick Fix: We can cure sharing problems most efficiently by drilling down through the possibilities in this order:

1. Ensure that the required services are running. On Windows systems, the server, TCP/IP NetBIOS Helper, workstation, and computer browser services all need to be running. On Linux machines, Samba is the primary component required to share with Windows systems.
2. Check your firewall(s). It's very common for a workstation's firewall to be configured to block file and printer sharing traffic, especially if a new antivirus package is installed that introduces its own firewall. Firewall issues can also exist at the hardware level, so ensure that routers or managed switches are passing share traffic within the subnet. Speaking of subnet….
3. Ensure all workstations are on the same subnet. This problem typically only appears on complex networks; however, even simple networks sometimes have static-IP equipment with an improperly configured subnet. The result is that external traffic will move about just fine, while internal traffic will hit unexpected roadblocks.
4. All Windows network adapters will need File and Printer Sharing for Microsoft Networks, Client for Microsoft Networks, and NetBIOS over TCP/IP enabled.
5. Once the above checks have passed, it's finally time to check the most likely culprit, permissions. There are multiple layers of access required, each with their own interface within the OS. Check for:

- Systems configured with the wrong workgroup or domain.
- Incorrectly configured HomeGroup.
- Network type set to Public.
- Incorrect NTFS permissions.

6. **Local Network is Unable to Connect to the internet**

This situation can either be intermittent or persistent. Often times, the most difficult aspect of dealing with any external network problem is finding the company responsible. And then tasking them to solve the issue, particularly with intermittent failures that are difficult to trace. It can sometimes be such a problem that organizations will have to switch internet providers in order to solve the issue.

The Quick Fix: A router and modem reboot is the first order of business. The tracert then utility can be used to identify communication breaks. It will clearly hiccup on the particular router hop that is causing the problem. Contact your ISP with your findings, providing screenshots as necessary.

The Preventive Measure: To avoid the finger-pointing that can prevent rapid resolution of external issues, do some research to ensure that you procure connectivity only from local Tier 1 providers. Other ISPs are more than happy to sell you service, however, they are simply piggybacking the Tier 1 connection, since they don't actually own the infrastructure in your area. The goal is to remove as many middle-men as possible, so that when (not if) you experience a problem, one phone call is all that is required to identify the issue and get technicians to work on it.

7. **Slow Internet Performance**

Slow performance is typically due to congestion, or sometimes poor-quality connections that have corroded or otherwise deteriorated. Congestion may not be directly related to bandwidth exhaustion, as a single overloaded port on a switch or router can diminish network performance. This can be especially true on leased lines where dedicated bandwidth is to be expected, but speed tests indicate the network is not reaching its rated potential.

The Quick Fix: Use speed test websites, conducting tests from geographically remote servers. This can pinpoint areas of congestion on the ISP's network. In the case of cable internet, the local network is shared amongst your neighbors, committing your ISP to a costly bandwidth upgrade when saturation occurs. Report your findings to your ISP so that they can take steps to resolve the issue. DNS servers are an often-overlooked aspect of internet performance. Using incorrect DNS servers can result in routing congestion or load balancing problems. While you should typically use your ISP's DNS settings whenever possible, they may actually be routing traffic through overloaded web caches. You can temporarily adjust your DNS settings to use OpenDNS instead.

The Preventive Measure: if internet performance is critical, you'll need to procure adequate connectivity. While cable internet may be inexpensive, you could be setting yourself up for frequent jeers from employees. A local DSL operator may offer improved reliability for a slightly higher cost, but for the most consistent performance, you may find that an expensive leased line is a requirement for your organization.

Assessment Task 5-1

Activity 1: Words to Know

Directions. Define the following terms on your own point of view (not exactly copied from the text).

1. Network

   _____

   _____

   _____

2. Netiquette

   _____

   _____

   _____

3. E-Commerce

   _____

   _____

   _____

4. E-Business

   _____

   _____

   _____

5. Internetwork

   _____

   _____

   _____

Assessment Task 5-2

Activity 2: Brush Up

Directions. Differentiate the following terms and give examples to support your answer.

1. LAN and MAN

   _____

   _____

   _____

2. WAN and VPN

   _____

   _____

   _____

3. DSL and Fiber

   _____

   _____

   _____

4. IPs and URL

   _____

   _____

   _____

5. Google Chrome and Apple Safari

   _____

   _____

   _____

Assessment Task 5-3

Activity 3: Sharpen Your Mind

Directions. Complete the table by identifying the Advantages and Disadvantages of the E-Commerce.

E-COMMERCE	
Advantage(s)	*Disadvantage(s)*

Summary

- Type of computer networks can categorize according to their size as well as their purpose.
- PAN is a computer network which generally consists of a computer, mobile, or personal digital assistant.
- LAN (local area network) is a group of computer and peripheral devices which are connected in a limited area.
- WAN (Wide Area Network) is another important computer network that which is spread across a large geographical area.
- A metropolitan area network or MAN is consisting of a computer network across an entire city, college campus, or a small region.
- WLAN is a wireless local area network that helps you to link single or multiple devices using. It uses wireless communication within a limited area like home, school, or office building.
- SAN is a storage area network is a type of network which allows consolidated, block-level data storage.
- System area network offers high-speed connection in server-to-server applications, storage area networks, and processor-to-processor applications.
- POLAN is a networking technology which helps you to integrate into structured cabling.
- Home network (HAN) is an always built using two or more interconnected computers to form a local area network (LAN) within the home
- Enterprise private network (EPN) networks are built and owned by businesses that want to securely connect various locations.
- Campus area network (CAN) is made up of an interconnection of LANs in a specific geographical area.
- A VPN is a private network which uses a public network to connect remote sites or users together.

References

Cloud, S. (2015). Networking and Data Analytics. Retrieved from: http://cloudcomputingnet.com/computer-network-components/

Deeksha, A. (2020). Advantages and Disadvantages of VPN (Virtual Private Network). Retrieved from: https://honestproscons.com/advantages-and-disadvantages-of-vpn/

Guru, J. (2013). Types of Computer Networks: LAN, MAN, WAN, VPN. Retrieved From: https://www.guru99.com/types-of-computer-network.html

MODULE 6
DATA SECURITY, DATA PRIVACY AND PROTECTION

Introduction

The ICT has changed the way people communicate and deal with other people. As a result, it affects businesses and organizations as ICT has become an essential part of our daily routines (e.g. work, business, to entertainment). According to Yulo (2018), digital data has become part of our personalities and identities. Through the internet and the social media, human relations are increasingly carried out on a global level and the local is shaped by events occurring at a far distance (Collste, 2008).

The main issue with the use of ICT products and services is that, it requires generation and collection of personal and sensitive information. Matter of fact, all companies handle some form of personal information – from employee to client to end-user data (Yulo, 2018). Additionally, organizations nowadays are producing more and more personal and sensitive data because of advances in technology advances, cheaper cost of storage, and the growing number of ICT users. The greater cause of concern here is that the volume of such data are not only being collected, they are also being shared by organizations (Accenture, 2009).

With the risk the organizations losing data and experiencing security breaches, the issue of data privacy and security should be everyone's concern.

Learning Outcomes

At the end of this period, students should be able to:

1. Learn about data computer security and privacy and why they matter;
2. Be familiar with the recent policy on data privacy and protection;
3. Be aware of the issues surrounding data privacy and data security and think of recommendations or solutions to such issues.

Lesson 1. COMPUTER and DATA SECURITY

Computer Viruses, Hacking and Spies

a) Computer viruses

A computer virus is a type of computer program that replicates itself once executed. What a virus does is that it modifies other computer programs and inserts its own code. The computer becomes “infected” if such replication succeeds (Bulik, 2014). Unlike a computer worm (which can run independently), a computer virus usually needs a host program where the virus writes its own code. When the program runs, the written virus program is executed first, causing infection and damage (Yeo, 2012).

b) Hacking

Hacking can be defined in different ways. It could mean compromising computer systems and networks, digital devices, or personal accounts. Intrinsically, hacking is not necessarily a malicious act. In other usage, hacking is the use of technology to successfully bypass a challenge (AVG, n.d.).

In the context of misusing computers to break the security of another computing system, hacking may include stealing of data, corrupting systems or files, or disrupting data-related activities (Techopedia, 2020).

c) Cyber spying or Cyber espionage

Cyber spying, or cyber espionage is the act or practice of gaining information and secrets without the consent, authorization, or awareness of the holder of such information using methods on the Internet, individual computers through proxy servers, or use of cracking techniques and malicious software. Information holders could be individuals, groups, governments, competitors, or enemies. People who commit cyber espionage do it for purposes ranging from personal, economic, political, or even military advantage (Techopedia, 2011).

Antivirus

a) **Antivirus software**

Antivirus is a computer program that helps protect your computer against malware and cybercriminals. Antivirus looks at data (e.g. websites, files, software, applications that travel over the network to your devices) and searches for known threats and monitors the behavior of all programs, flagging suspicious behavior (Johansen, 2019).

b) **Virus versus malware**

The terms “malware” and “virus” are often used interchangeably but they are technically different.

- **Malware** is an umbrella term for any type of malicious software written specifically to infect and harm the host system or its user (Avast, n.d.). Malware is a catch-all term for any type of malicious software, regardless of how it works, its intent, or how it’s distributed (McAfee, n.d.).
- A **computer virus** is a specific type of malware that self-replicates by inserting its code into other programs (McAfee, .d.). All viruses are malware, but not all malware is a virus (Avast, n.d.).

Types of Hackers

Hackers can be classified into three broad groups, based on the legality of their activities. Here are the three types of hackers:

a) **Black hat hackers.**

These are hackers who breach a computer system to steal information, modify records, or carry out other illegal purposes. A black hat hacker looks for vulnerability in a piece of software and take advantage of that flaw (AVG, n.d.).

b) **White hat and ethical hackers**

Unlike their black-hat counterpart, white hat hackers do all their hacking out in the open. White hat hackers are even hired to intentionally hack a company’s software

and systems to detect vulnerabilities or flaws in security. White hat hackers may be working as freelancers, contractors, or in-house at large organizations (AVG, n.d.).

- **Penetration testing –** refers to the deliberate hacking technique
- **Ethical hacking** – refers to the practice done by white hat hackers
- Ethical hacking tests a system's resistance to real-life attacks and help identify areas which need additional cybersecurity. The process prevents the real threats posed by black hat hackers.
- White hat hackers are **authorized** to hack systems to test resiliency and spot weakness (Techopedia, 2020).

c) **Grey hat hackers**

Grey hat hackers are not explicitly ethical hackers nor are they involved in criminal acts. Unlike ethical hackers, the grey hat hackers tend to hack first before asking for permission later. Grey hat hackers are unauthorized but are not acting in a malicious manner (AVG, n.d.).

- Some grey hats identify a system/security flaw and then sell the solution.
- Others use hacking for activism, by publicly exposing vulnerabilities so that the targeted company has no choice but to fix them.
- While grey hat hacking is for positive intent, security hacking without prior consent is **still against the law.**

Types of Cybercriminals

According to Techopedia (2020), there are three types of cybercriminals: 1) the lone wolf, 2) state-sponsored agents, and c) corporate actors.

1. **Lone Wolfs.** These are actors who act alone who may seek to test their computing skills and cause random damage.
2. **State-Sponsored Agents.** These are often security and signals specialists who work in classified settings to collect data (e.g. commercial or government), or sometimes to carry on cyber warfare.
3. **Corporate Actors.** They are said to be rare, but one who takes the long-standing concept of corporate espionage into the digital realm. **Corporate espionage** (also

called industrial espionage, economic espionage, or corporate spying) is the practice of using espionage techniques for commercial or financial purposes. In corporate espionage, companies gather intelligence on their rivals just like nation-states do. Sometimes its legal, but industrial espionage can easily slip over the line into criminality (Fruhlinger, 2018).

Corporate espionage

Here are some of the techniques that fall under the umbrella of industrial espionage (Fruhlinger, 2018):

1. Trespassing onto the property of the competitor and gain access to their files without permission;
2. Pretending to be an employee to learn the company’s secrets;
3. Wiretapping the competitors;
4. Hacking into a competitor's computers
5. Attacking a competitor's website with malware

Lesson 2. DEFINITIONS AND VIEWS ON DATA

2.1 Data is the New Oil

Data is not just a concern of scientists nowadays. More than ever, data has become central and essential for businesses and industries focused on technology, infrastructure, finance, manufacturing, insurance, and energy. These entities treat data as a form of capital (Sadowski, 2019).

"The world's most valuable resource is no longer oil, but data" (The Economist, 2017). According to Mavuduru (2020), data is like oil in three ways: a) it should be refined to be valuable, b) data analytics is like combustion engine, and c) data, like oil, requires infrastructure.

Organizations today are generating more data than ever (Accenture, 2009). Industries such as those focused on infrastructure, manufacturing, insurance, finance, energy, and technology are now treating data as a form of capital (Sadowski, 2019). Along with the increase in the volume of collected data is the rise in the probability of compromising such data (Accenture, 2009).

2.2 Why Data Protection is Needed

Since the 1960s and the expansion of ICT capabilities, businesses and governments have been collecting and storing personal data in databases. Whenever we use online services (shopping, email, pay our taxes, enter a contract or service requests), we are giving away over some of our personal data and information and these are being collected by companies and agencies with or without our knowledge (Privacy International, 2018).

Along with the increase in the volume of collected data is the increased risk of compromising the data various ways. Data compromise on the part of organizations could lead to financial costs for damage repair, fines, decline in share price, and overall mistrust from the customers (Accenture, 2009).

Lesson 3. THE DATA PRIVACY ACT

3. 1 the Data Privacy Act of 2012 (RA 10173)

The Data Privacy Act (DPA) of 2012, otherwise known as the Republic Act No. 10173, is a law that seeks to protect all forms of information (i.e. private, personal, sensitive) "to protect the fundamental human right of privacy, of communication while ensuring free flow of information to promote innovation and growth" (Wall, 2017).

The DPA applies to any applies to any process of personal data by anyone in government or private sectors (Yulo, 2018). The law has extraterritorial application. This means the law applies not only to businesses with offices in the Philippines, it also applies to the processing of the personal information of Filipino citizens abroad, except for personal information that were collected legally esp. for companies that offer cloud services (Wall, 2017).

3.2 Highlights and Importance of the Data Privacy Act

Vast amounts of personal information from the Philippines are freely circulating the Web. Filipinos are said to be online for about 10 hours and 2 minutes every day – the highest in the world. The Philippines also tops social media use for the fourth straight year (as cited by ECCI International (n.d.).

The right to privacy is a fundamental right. The fundamental aspect of this right is the protection of individuals' data (Privacy International, 2018). In this connection, the DPA aims to address the 21st century issues and crimes as follows (National Privacy Commission, n.d.):

1. To protect the privacy of individuals at the same time ensuring the free flow of information to promote innovation and growth;
2. To regulate the collection, recording, organization, storage, updating or modification, retrieval, consultation, use, consolidation, blocking, erasure or destruction of personal data; and

3. To ensure that the Philippines complies with international standards set for data protection through National Privacy Commission (NPC).

3.3 Data Subjects

Data subjects refer to identified or identifiable person to whom Personal Data relates (Law Insider, n.d.). The data subjects are individuals sharing their personal information (Yulo, 2018). The DPA states that consent is required prior to the collection of their personal data and the data subjects must be fully informed of several factors of the data collecting process. This means that companies must ensure that their methods of data collection methods must be shared with data subjects, including a breach of security, should there be any (Yulo, 2018). When obtaining consent, the data subjects must be informed about the extent and purpose of processing, which includes profiling, processing for direct marketing, and data sharing (Wall, 2018).

3.4 Personal Information

Personal information refers to all types of personal information (ECCI International, n.d.), whether recorded in a material form or not, from which the identity of an individual is apparent or can be reasonably and directly ascertained by the entity holding the information, or when put together with other information would directly and certainly identify an individual (Republic Act. No. 10173).

Sensitive personal information refers to (1) an individual's race, ethnic origin, marital status, age, color, and religious, philosophical or political affiliations; (2) an individual's health, education, genetic or sexual life of a person, or to any proceeding for any offense committed or alleged to have been committed by such person, the disposal of such proceedings, or the sentence of any court in such proceedings; (3) government-issued data, such as social security numbers, licenses or its denials, suspension or revocation, and tax returns; and (4) Information declared classified by an executive order or an act of Congress (Republic Act. No. 10173).

3.5 National Privacy Commission

The National Privacy Commission (NPC) is an independent body which was formed under the Data Privacy Act of 2012 and is attached to the Philippines' Department of Information and Communications Technology (Republic Act No. 10173).

The NPC oversees the administration and implementation of the DPA. The NPC is tasked to monitor and ensure compliance of the Philippines with international standards for personal data protection. The major functions of the NPC are as follows (ECCI International, n.d.):

a) **Advisory.** The NPC serves as advisory body on matters related to personal data protection.
b) **Public education**. The NPC shall launch initiatives to educate the public about data privacy, data protection and fair information rights and responsibilities.
c) **Rule-making and enforcement.**
d) **Compliance and monitoring**. The body has compliance and monitoring functions to ensure personal information controllers comply with the law. It is also tasked to manage the registration of personal data processing systems.
e) **Complaints and investigations**.

3.6 Personal Information Controller (PIC)

A **personal information controller** is an individual or institution, or any other body who controls the processing of personal data, or instructs another to process personal data on its behalf (ECCI International, n.d.).

3.7 Personal Information Processor (PIP)

Personal information processor or “PIP” refers to any natural or juridical person or any other body to whom a PIC may outsource or instruct the processing of personal data pertaining to a data subject (Republic Act. No. 10173).

3.8 Data Protection Officer (DPO)

Data Protection Officers (DPO) are appointed by companies to ensure that their data collection methods are flawless, to regulate the handling of information, as well as conduct routine assessments to ensure quality data protection (Yulo, 2018).

Lesson 4. Case Examples, Issues and Concerns

4.1 COVID-19 and Data Protection Compliance

Dealing with the COVID-19 pandemic required tracking and tracing of both suspected and confirmed cases of COVID-19. This is the subject of the Republic Act 11332 which mandates the Department of Health (DOH) the collection and processing of COVID-19 related data of both personal and sensitive information (DOH-NPC, 2020). This has raised the issue on right to privacy esp. on the collection and disclosure of patient information (Baker McKenzie, 2020).

The NPC has since advised the government to seek the balance between public health interests and individual privacy amidst the increasing the number of COVID-19 cases. While the DPA does not prevent the government and health entities from processing personal and sensitive information, it must only be done when it contributes to the fulfillment of their mandates during a public health emergency that the unnecessarily disclosure of personal information may have serious consequences (IAPP, 2020).

In response to the growing privacy concerns raised on the COVID-19 health situation, and in upholding RA 11332 and RA 10173, the DOH and the NPC have issued the Joint Memorandum Circular 2020-0002 which details the guidelines on integrating data protection and privacy principles in the collection, processing, and disclosure of COVID-19-related data (DOH-NPC, 2020).

4.2 Examples of Breaches and Live Cases

a) COMELeak

The 2016 hacking of the Commission of Elections (Comelec) website is said to be the biggest private data leaks in Philippine history (Valenzuela, 2020). This left 55 millions of registered Filipino voters at risk, surpassing the 2015 US Office of Personnel Management hack that leaked personal data of 20 million Americans (Malig, 2016). Chi (2016) even estimates the

number at 70 million people. This hacking is said to be the biggest breach involving government-held data (Jacob, 2018).

The kinds of data that were believed to be compromised were personal data such as full names, birth dates, their parents' names, some passport details and home and e-mail addresses (Malasig, 2018). Other information include passport details of some 1.3 million OFW voters, fingerprints of 15.8 million Filipinos (Malig, 2016). The hacker group Anonymous Philippines claimed responsibility for the attack stating that their purpose was to show vulnerabilities in the COMELEC's system, highlighting that the upcoming elections will use automated voting machines. Days later, the entire database was said to have been leaked online by the second hacker group who identifies as LulzSec Philippines (Chi, 2016).

b) The Facebook–Cambridge Analytical Data Scandal

The Facebook–Cambridge Analytical data scandal involved the harvesting of the personal data of up to 87 million of Facebook profiles without their consent (Criddle, 2020). The harvested data came from a personality quiz titled "thisisyourdigitallife" which around 270,000 people were paid to take. The quiz takers were unaware that not their data were collected but also their friends' profiles as well (Ma and Gilbert, 2019).

The collected data were said to have been used primarily for political advertising (Chan, 2020) and used them to build a software program to predict and influence choices at the ballot box (Cadwalladr and Graham-Harrison, 2018). Mass legal action has been launched against Facebook for misuse of information (Criddle, 2020).

Assessment Task

1. Look for three (3) examples or cases **[10 points each]** that involved hacking, data breach, or data leaks. Briefly describe what happened. Some details that you can include are: (a) where did it happen, (b) when did it occur, (c) who were involved. Use your own words. [**30 points Total**]

2. As a student, in what ways does data security and data privacy affect you personally? [**20 points**]

Summary

The use of ICT products and services are increasingly becoming more pervasive as the computer, the internet, and smartphones have evolved to become an everyday necessity. We now spend a significant portion of our daily lives online for purposes such work and entertainment. Because of this cyber-presence, our constant exposure to online content, and our use of digital platforms and services, we are also continually exposing ourselves and our privacy to risks.

Since both the government and private organizations collect our information in several ways, we as citizens and consumers must assert for stronger data protection practices or call for effective legislation to minimize surveillance and data exploitation. The good thing is that we, as data subjects, are somehow protected by the RA 10173. The law requires organizations who deal with our personal details, whereabouts, and preferences are duty bound to observe and respect our data privacy rights.

It must be recognized however that data privacy and protection is everyone's concern and private citizens like us have a role in protecting our data and privacy rights.

References

Accenture. (2009). How Global Organizations Approach the Challenge of Protecting Personal Data. [Ponemon Institute]. Accessed from https://bit.ly/3bITeJY on 2021 February 20.

Avast. (n.d.). Malware vs. Viruses: What's the Difference? Accessed from https://www.avast.com/c-malware-vs-virus on 2021 February 20.

AVG. (n.d.). What Is Hacking? Everything You Need to Know. Accessed from https://bit.ly/3aPUADe, 2021 February 24.

Baker McKenzie. (2020, June 3). Protecting patients' privacy in a pandemic: latest guidance from the Philippines. Accessed from https://bit.ly/3dQTqt7 on 2021 February 27.

Bulik, M. (2014, August 6). "The Internet comes down with a virus". The New York Times. Accessed from https://nyti.ms/2Nxbn5e on 2021 February 24.

Cadwalladr, C. and E. Graham-Harrison. (2018, March 17). Revealed: 50 million Facebook profiles harvested for Cambridge Analytical in major data breach. [The Guardian]. Accessed from https://bit.ly/3bI8ff8 on 2021 February 27.

Chi, L. (2016, April 11). Philippines elections hack 'leaks voter data'. [BBS News]. Accessed from https://bbc.in/3dSOGn4 on 2021 February 27.

Chan, R. (2019, Cotober 6). The Cambridge Analytical whistleblower explains how the firm used Facebook data to sway elections. [Business Insider]. Accessed from https://bit.ly/3q1RlNz on 2021 February 10.

Collste, G. (2008). Global ICT-ethics: the case of privacy. Journal of Information, Communication & Ethics in Society Vol. 6 No. 1, 2008. pp. 76-87.

Criddle, C. (2020, October 28). Facebook sued over Cambridge Analytical data scandal. [BBC News]. Accessed from https://bbc.in/2O79FHG on 2021 February 27.

DOH-NPC. (2020, April 24). DOH-NPC Joint Memorandum Circular 2020-0002. Accessed from https://bit.ly/3sw6AzZ on 2021 February 27.

ECCI International. (n.d.). A Summary of RA No. 10173 or the Data Privacy Act of 2012. Accessed from https://bit.ly/3sDC8Eh on 2021 February 27.

The Economist. (2017, May 06). The world's most valuable resource is no longer oil, but data. Accessed from https://econ.st/37NG4dx on 2021 February 25.

Fruhlinger, J. (2018, July 2). What is corporate espionage? Inside the murky world of private spying. [CSO-ASEAN]. Accessed from https://bit.ly/3q1RyjP on 2021 February 25.

IAPP. (2020, March 19). Philippines NPC issues privacy reminder during COVID-19 outbreak. Accessed from https://bit.ly/3dLVi6C on 2021 February 27.

Jacob, J. (2018, May 09). Revisiting ComeLeak, by the numbers. [Ateneo de Manila University website]. Accessed from https://bit.ly/2O1nqYy on 2021 February 27.

Johansen, A. G. (2019, February 22). What is antivirus software? Antivirus definition. [Norton]. Accessed from https://nr.tn/3pX2sqZ on 2021 February 25.

Law Insider. (n.d.). Data Subject definition. [Law Insider]. Accessed from https://bit.ly/2ZTeUNN on 2021 February 27.

Ma, A. and B. Gilbert. (2019, August 24). Facebook understood how dangerous the Trump-linked data firm Cambridge Analytical could be much earlier than it previously said. [Business Insider]. Accessed from https://bit.ly/3bLtFYM on 2021 February 27.

Malasig, J. (2018, April 17). Facebook data breach or Comeleak: Which could have a bigger impact? [Interaksyon website]. Accessed from https://bit.ly/3dQLUi1 on 2021 February 27.

Malig, J. (2016, April 07). Comelec hacking threatens security of voters: Trend Micro. [ABS-CBN News]. Accessed from https://bit.ly/2NNEZvj on 2021 February 27.

Mavuduru, A. (2020, Dec 12). Is Data Really the New Oil in the 21st Century? Exploring the strengths and limitations of this metaphor in the information age. [Towards Data Science]. Accessed from https://bit.ly/3aZ6vPu on 2021 February 25.

McAfee. (n.d.). What Is the Difference Between Malware and a Virus? [McAfee website] Accessed from https://bit.ly/3099E9f on 2021 February 27

National Privacy Commission. (n.d.) A brief primer on Republic Act 10173. [GovPH]. Accessed from https://bit.ly/3q3J2Rk on 2021 February 27.

Privacy International. (2018 August). The Keys to Data Protection: A Guide for Policy Engagement on Data Protection. UK Registered Charity No. 1147471

Republic Act 10173 – Data Privacy Act of 2012. (n.d.). Retrieved from https://bit.ly/3qW4adw on 2021 February 27.

Techopedia. (2011, August 18). Cyberspying. Accessed from https://bit.ly/2ZHxl8c on 2021 February 24.

Techopedia. (2020, March 27). Hacking. Accessed from https://bit.ly/3aLW959 on 2021 February 24.

UNHCR. (2015). Policy on the Protection of Personal Data of Persons of Concern to UNHCR.

Valenzuela, N.G. (2020, February 29). 'Comeleak' hacker cleared of cybercrime charges. Accessed from https://bit.ly/3kv1LnN on 2021 February 25

Wall, A. (2017, April 27). Summary: Philippines Data Privacy Act and implementing regulations. Accessed from https://bit.ly/3uA8EZz on 2021 February 25.

Yeo, Sang-Soo. (2012). Computer science and its applications: CSA 2012, Jeju, Korea, 22-25.11.2012. Springer. p. 515. ISBN 978-94-007-5699-1. OCLC 897634290.

Yulo, S. (2018, April 11). The Data Privacy Act (RA 10173): Here's What You Need to Know. [Sprout Solutions]. Accessed from https://bit.ly/3qULJWF on 2021 February 27.

MODULE 7
POLITICAL, ECONOMIC AND SOCIETAL IMPACTS AND CHANGE

Introduction

Information and communications technology (ICT) and its many products have become an integral part of our modern everyday lives. We use our smartphones for remote conversations via calls and text messaging along with personal computers for online communications. Just the same we use the internet for work, business, and entertainment. Thoughts and ideas have been expressed ubiquitously through the internet. The ICT affects us on a personal level in many ways.

Looking at a much bigger picture, the ICT powers the growth of our economies and industries by influencing corporate growth, investments, global distribution of jobs. Through the accessibility of useful information, the ICT also stimulates competition. Much has changed since the first products of ICT were conceptualized and introduced and it has now become a fundamental force for global change (The Conference Board, 2011). The enhanced exchange of information through the ICT has tremendous impacts on the global politics and economy through faster transactions and the real-time availability of data and information from anywhere in the world.

The Internet has virtually removed the barriers of physical distance by connecting people across geographical boundaries. This process has impacts on the environment, cultures, political systems, economies, and on well-beings of people from societies around the world.

Learning Outcomes

At the end of this module, students should be able to appreciate how information and communications technology has influenced and changed economics, politics, and societies around the globe.

Lesson 1. Economic Impacts of ICT

Positive Impacts of ICT on Economics

In business activities, the role of ICT can be summarized into two: production and consumption. Firms can benefit from the manufacture of ICT hardware and software by producing it themselves, by consuming the said ICT goods, or through the new economies created as a result of the public adoption of said ICT goods/services (Draca, Martin & Sanchis-Guarner, 2018).

On the economy, the impact of ICT can be analyzed from three perspectives: supply, demand, and distribution (The Conference Board, 2011). The ICT affects the economy in three ways (OECD, 2001):

1. ICT investment contributes to capital deepening. Capital deepening means higher output per worker. ICT helps increase labor productivity.
2. Technological progress through production of ICT goods and services contribute to growth in the efficiency of capital and labor growth in the ICT sector; and
3. The use of ICT leads to efficiency gains in individual firms or network/spillover effects.

As cited in Toader, Firtescu, Roman & Anton (2018), the rapid expansion of ICT influences economic growth in the following ways:

1. Quick and easy access to information and knowledge.
2. Faster and better communication for companies which leads to reduced production costs and improved productivity.
3. ICT allows access to new markets. This leads to lowered capital costs because of financial markets become more efficient, there is also regional discrepancies in incomes and productivity, and access to human capital is made possible through tele-networking.
4. The Internet promotes the sustainable development of entrepreneurship and small and micro businesses by reducing the difficulty of financing them by mitigating information asymmetry and reducing agency cost.

Negative Impacts of ICT on Economics

The proliferation of computers and related technologies into the labor market has been perceived to cause displacement of jobs through automation esp. for those working in the so-called routine jobs (Draca et al., 2018). Routine jobs include those involving manual operations that can be replaced by automation, e.g. robots instead of people on assembly lines. Some examples of jobs that are affected by the ICT, computing technology, artificial intelligence (AI) and robotics-based innovations are:

1. **Job exports.**

   Due to ICT and the internet, work such as data processing can be outsourced from overseas esp. where services are cheaper.

2. **Automation will lead to job loss.**

   Examples: Workers on supermarket checkouts can now serve more customers per hour through bar-code scanners. Gone are days when cashiers need assistance from other workers by manually checking the price. ICT can also give rise to autonomous vehicles. Self-driving cars may not be needing the services of drivers anymore.

3. **Artificial intelligence threatens to replace people.**

   Customer service representatives maybe threatened by AI technologies. In call centers for instance, a large number of calls are usually basic queries that can be dealt with by an AI with usual answers such as 'Have you tried turning your computer off and on?'

Lesson 2. Political Impacts of ICT

E-participation

Democratic participation of citizens via ICT-based systems is called **e-participation.** In this day and age, democratic participation can now be done digitally and online by means of e-participation. Several factors can affect the level of democratic activity: a) age of citizens, b) gender, c) education, d) wealth, e) race (Jafarkarimi, Sim, Saadatdoost & Hee, 2014). Emer and Kunst (2018) argue that the use of social media and political participation are positively related.

ICT and Democracy

Jafarkarimi et al. (2014) discussed the role of ICT and the social media on democratic activities, their impacts on governments' decision making, and the change in citizens' participation in democratic activities.

According to Tsagarousianou (1999), ICTs can potentially contribute to the democratic process through three different ways:

1. **Obtaining information**.
   ICTs provide information about the government through websites and online news coverage. Most of them support a real-time dissemination of political events.

2. **Engaging in deliberation.**
   ICTs bridge the communication between citizens and their representatives.

3. **Participating in decision making.**
   The most familiar ways in which citizens participate in politics is during elections. ICTs could help ease the voting process through electronic voting by making the process available remotely and speeding up the counting of votes.

Positive Impacts of ICT on Politics

Nam (2011) believes that the internet can contribute to the improvement of equality and inclusiveness of political participation by reinforcing the existing patterns in offline political participation and mobilizing a new pattern for online political participation.

Social media also plays an important role in empowerment, mobilization, shaping opinions, and influencing changes (Mourtada and Salem, 2011). In Africa for example (Dobra, 2012), increased online participation by citizens have led to:

1. ICT promotes culture of openness and increased transparency.
2. ICT-based forums as sources of publicness and political accountability.
3. ICT as a platform for citizen interaction and criticism of state operations.
4. ICT promotes sense of trust and reciprocity between citizens and the state.

Negative Impacts of ICT on Politics

Dobra (2012) cites an issue that ICT is seen by most political elites, policy makers, and international organizations as a device for instigating a democratic revolution without seeking to assess the social and human consequences of that belief.

Utilizing ICT for democratic processes is not without risks. Jafarkarimi et al. (2014) cites the following list of disadvantages:

1. ICT can encourage populist participation.
2. ICT may reduce the possibility of collective action.
3. ICT may erode social capital and community ties and civil discussions.

Lesson 3. Social Impacts of ICT

ICT has been a significant part of our daily lives. We get news from online platforms. We connect with friends and relatives through social media and with our colleagues through e-mails and call, or video conferences (Arugu and Chigozie, 2016).

Impacts of ICT on Relationships

The interactive nature of modern information and the ICT affects relationships in many ways. In this context, the effects of ICT can be any of the following (The Conference Board, 2011):

1) **Strengthening of existing relationships**
   Relationships are maintained and reinforced through communications via social media. Unlike before, physical distance and geographic isolation are no longer obstacles for retaining the networks and preserving the relationships.
2) **Alteration of existing relationships**
   Through the social media, people can comfortably disclose their deeply held beliefs and stand on issues. In some cases, such beliefs may be in opposition with their friends and acquaintances. Social media helps deepen familiarity and acquaintance.
3) **Creation of entirely new relationships**
   Through social media, people can establish new connections with strangers.

Positive Impacts of ICT on People

The following positive impacts of ICT on people are excerpts from Bosamia (2018):

1. **Daily Routine Management.**
   Daily schedules can be stored and managed using ICT gadgets.
2. **Online banking and shopping**.
   Purchasing needed items are now convenient. ICT saves time.
3. **Maintaining contacts and relationships**.
   Through the Internet, people can connect via e-mail, social networking sites and phones. Discussion groups' services are available for communicating with many people at the same time.
4. **Global Awareness**.

Information can be accessed anywhere and anytime. News and opinions shared by people all around the globe promotes global awareness on cultures, religions, ideologies and lifestyles. This creates an environment of openness and mutual respect.

5. **Easier Communication and Information Sharing**.

   Communication has become quicker and instant. A myriad of platforms are now available: e-mail, teleconferencing, and video-conferencing. Unlike before, people can now have remote and virtual face-to-face interactions.

6. **Entertainment.**

   ICT provides multitude of options for entertainment: playing games, watching videos, listening to songs, etc.

7. **Education.**

   New ways of education and learning are now possible through ICT: e-learning, distance learning, online materials, virtual learning, and interactive multi-media.

8. **Self-Employment.**

   ICT devices makes jobs easier by allowing people to create business shops virtually without the heavy overheads of place, purchase items and sales in market and make the advertising campaigns use of multimedia applications.

9. **Health Literacy.**

   People now have access to useful and high-quality health information about diseases and medicines through the internet and social media. A good example is useful information about COVID-19, prevention, and the current status of the pandemic.

Negative Impacts of ICT on People

1. **Isolation of Individuals.**

   ICTs can reinforce the isolation of individuals and groups who lack access and skills in ways that hamper not only their economic development, but also their understanding of and participation in society and culture (The Conference Board, 2011).

2. **Reduced face-to-face interaction.** This results from online communication rather than real conversations (Bosamia, 2018).

3. **Social disconnect.** Although the internet has bridged people who are the physically distant, emotional distance is another issue (Bosamia, 2018).

4. **The digital divide.**

Digital divide refers to a gap between individuals or countries based on unequal access to ICT. The impact of such unequal access on individuals ranges from inconvenience (i.e. paying the bills in person rather than online) to a more serious outcomes (i.e. disadvantage in workplaces due to unfamiliarity with ICT). Unequal access in economies can be in the form of lack of ICT infrastructure (OECD, 2008).

5. **Job loss.**

   This is primarily a social issue as it is an economic concern. Some manual operations like simple counting of money can now be done by machines. Job losses may occur because of automation (Bosamia, 2013).

   a. Robots could replace people on an assembly line.
   b. Data processing jobs can be outsourced from other countries where services are cheaper.
   c. Work that requires multiple workers are being replaced by a smaller number people due to computerized tools.

6. **Effects on children.**

   Children exposed to various forms of information may have impacts on their religious values, norms, and social boundaries. Among the issues widely discussed are erosion of individual self-worth and patriotic spirits, increase in immoral activities, and fragile family institution. This can become a challenge for educators and teachers (Salleh and Othman, 2013).

7. **Reduced physical activity.** Too much time spend on ICT gadgets could mean extended periods of time in sitting position. This could lead to health problems for adults. For children, mobile games and watching cartoons may lead to less outside physical activities and reduced interaction with other children. Sedentary lifestyles mean less physical activities thus increased health risks (Bosamia, 2013).

8. **Security/loss of privacy.**

   With no real national boundaries, the legal implications of the Internet and the World Wide Web are immense. Thorny issues such as intellectual property protection, privacy, security, data protection, electronic payments and currency, and wide-ranging consumer protection issues must be addressed in national legislation and regional

strategies; each with tremendous social and economic implications (Cogburn and Adeya, 1999).

Assessment Task

1. As a student, in what ways does ICT help you? How does ICT affect you in positive ways? List 5 examples. [**20 points**]

2. As a student, in what ways does ICT affect you negatively? List 5 examples. [**20 points**].

3. List 5 things that you learned in this Module. [**10 points**]

Summary

The ICT has greatly advanced our speedy access to information regardless of our geographical location and the origin of such information. We now have access to real-time news and information from anywhere in the world, any time of the day. Because of the ICT, our sense of politics, society, and economics now goes beyond the physical barriers of physical geography and country boundaries.

Since the launching of personal computers in 1982, the development of the Internet in the 1990s, and the rise of mobile devices in the 2000s, the public has been utilizing the many products of the ICT for both work and personal uses. The use of ICT products and services are increasingly becoming more pervasive as our smartphones are now evolving to become an everyday necessity. Our cyber-presence and exposure to online content have impacts on our well-being, on how we perform our roles in our society, which also have direct and indirect implications on the economy, politics, and the global virtual community.

References

Arugu, L.O. and C.F. Chigozie. (2016). Information and Communication Technology (ICT) Application in Social and Political System. European Journal of Research in Social Sciences Vol. 4 No. 1, 2016. ISSN 2056-5429.

Bosamia, M. (2018). Positive and Negative Impacts of Information and Communication Technology in our Everyday Life Conference Pape, December 2013. https://www.researchgate.net/publication/325570282

Cogburn, D.L. and C.N. Adeya. (1999). Globalization and the Information Economy: Challenges and Opportunities for Africa. Working paper for the African Development Forum '99. 24-28 October 1999, United Nations Conference Centre. United Nations Economic Commission for Africa

Dobra, A. (2012). The Democratic Impact of ICT in Africa. Africa Spectrum 1/2012: 73-88.

Draca, M., R. Martin & R. Sanchis-Guarner. (2018). The Evolving Role of ICT in the Economy A Report by LSE Consulting for Huawei. LSE Consulting, LSE Enterprise Limited, London School of Economics and Political Science.

Emer, M. and M. Kunst. (2018). “Digital Citizenship” Revisited: The Impact of ICTs on Citizens’ Political Communication Beyond the Western State. International Journal of Communication 12(2018), 2191–2211 1932–8036/20180005. Available at http://ijoc.org.

Jafarkarimi, H., Sim, A., Saadatdoost, R. & Hee, J.M. (2014). The Impact of ICT on Reinforcing Citizens’ Role in Government Decision Making. International Journal of Emerging Technology and Advanced Engineering. ISSN 2250-2459, ISO 9001:2008 Certified Journal, Volume 4, Issue 1, January 2014).

Joseph, J. (2006). The Media and Globalisation. Journal of Communication Studies. India: Makhanlal Chaturvedi National University of Journalism.Vol.5, No.I-2. January-June.

Mourtada, R. and F. Salem (2011). The Role of Social Media in Arab Women's Empowerment. Dubai, Dubai School of Government, 1: 26.

Nam, T. (2011). "Dual effects of the internet on political activism: Reinforcing and mobilizing." Government Information Quarterly.

OECD. (2008). Measuring the Impacts of ICT Using Official Statistics. OECD Digital Economy Papers, No. 136, OECD Publishing, Paris. http://dx.doi.org/10.1787/230662252525

The Conference Board. (2011). The Linked World: How ICT is Transforming Societies, Cultures, and Economies. Research Report R-1476-11-RR. ISBN NO. 0-8237-1012-2.

Toader, E., B. N. Firtescu, A. Roman and S. G. Anton. (2018). Impact of Information and Communication Technology Infrastructure on Economic Growth: An Empirical Assessment for the EU Countries. Sustainability 2018, 10, 3750; doi: 10.3390/su10103750

Tsagarousianou R. (1999) "Electronic democracy: Rhetoric and reality", Communications; The European Journal of Communication Research, Vol 24, No.2) pp 189-208.

United Nations. (2004). Meeting the Challenges in an Era of Globalization by Strengthening Regional Development Cooperation. ECONOMIC AND SOCIAL COMMISSION FOR ASIA AND THE PACIFIC. UNITED NATIONS, New York. SBN: 92-1-120378-3